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**Seminar in Topics in Labor Economics and the Economics of
Education**

Did Remote Work Narrow the Motherhood Penalty?

**Evidence on the Intensive Margin from Israeli Labor Force
Survey Microdata, 2017–2023**

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Abstract

We ask whether the post-pandemic rise of work from home (WFH) narrowed the motherhood penalty in usual weekly hours among employed women in Israel. We use Central Bureau of Statistics (CBS) Labor Force Survey microdata on women aged 25–59 in 2017–2019 and 2021–2023 to estimate difference-in-differences (DiD) and triple-differences (DDD) models of usual weekly hours, with an Israeli-calibrated occupational WFH-exposure index as the third difference. Employed mothers’ hours did not change on average (0.23 hours, SE 0.18), but they rose where the work can be done from home: the triple interaction is 3.22 hours per unit of exposure (SE 1.02), or 1.71 hours a week between the mean exposures of the most and least teleworkable quartiles of occupations. Because exposure varies across only forty occupations we report three inference procedures: the analytic p -value is 0.003, the permutation p -value 0.032 and the wild cluster bootstrap p -value 0.056. The gradient survives a saturated fixed-effects specification, a selection correction and the removal of any single occupation; it is largest for mothers of children under five, and fathers show the opposite gradient. The employment margin is estimated too imprecisely to support a comparable conclusion. The result is about *which* mothers benefit from remote work rather than about mothers as a group.

1 Introduction

The shift to remote work is a rare natural experiment in workplace flexibility. Unlike the acute disruption of 2020, work from home has proven to be a persistent feature of the post-pandemic labor market (Barrero, Bloom and Davis, 2021), and flexibility has been identified as the last major driver of the remaining gender pay gap (Goldin, 2014). Temporal and locational flexibility is a good mothers pay for, in lower wages or slower promotion, so a shock that lowers the cost of supplying it should be detectable in mothers’ labor supply, and for mothers already employed most directly in how many hours they supply. This paper asks whether the post-COVID shift to remote work altered the motherhood penalty in *usual weekly work hours* among employed women in Israel, using Central Bureau of Statistics (CBS) Labor Force Survey microdata for 2017–2023.

The question has been asked once before, for the United States, where work from home reduced the motherhood penalty on the extensive margin and on income (Harrington and Kahn, 2025): a 10% rise in WFH raises mothers’ employment by roughly 0.78 percentage points relative to other women, and employed mothers’ incomes rise by about 1.3%. We ask it of a different country and a different margin. Israel combines a productive, internationally integrated labor market with the OECD’s highest total fertility rate, 2.87 children per woman in 2024 against an OECD average of 1.40 (OECD, 2025), so an unusually large share of its female workforce is at any time the mother of a young child, and an intensive-margin response to flexibility has first-order consequences for aggregate

labor input. Between 13% and 16% of employed Israeli women aged 25–59 usually worked from home in 2021–2023, the first years in which the survey asked. We also estimate the employment margin, but for reasons of statistical power documented in Section 6.6 we treat it as secondary.

Our design exploits the pandemic as the moment at which remote work became a mainstream practice. We compare mothers and childless women across a pre-period of 2017–2019 and a post-period of 2021–2023, excluding 2020 (Section 3). A difference-in-differences on this contrast estimates the change in the hours penalty. A triple-differences model that interacts it with an occupation’s WFH exposure tests whether remote work is the *mechanism*, by asking whether the change is larger in jobs that can actually be done from home. Its identifying assumption concerns how the motherhood gap *within* occupations would have evolved across exposure levels absent the shift; pre-period event studies probe that assumption rather than verify it.

Three findings emerge. Mothers’ hours did not rise on average: the DiD is 0.23 hours (SE 0.18), a reasonably precise null. The penalty did narrow where the work can be done from home: the triple interaction is 3.22 additional weekly hours per unit of exposure, or 1.71 hours between the mean exposures of the top and bottom quartiles of occupations, the same order as the 1.74-hour intensive-margin penalty itself, while the implied change at zero exposure is if anything negative and the response is concentrated in the most teleworkable quartile. And the gradient behaves as a flexibility mechanism should: it is largest for mothers of children under five and declines with the age of the youngest child, and fathers in the same occupations show a gradient of the opposite sign. Because the exposure score varies across only forty occupations, we report analytic, permutation and wild cluster bootstrap inference side by side, and the strength of the statistical conclusion depends on which is used (Section 6.2). The estimate survives specification, selection, outcome-coding, leave-one-occupation-out and calibration checks and a test of sorting across occupations, although the one exposure measure built entirely from pre-treatment information yields a weaker, insignificant estimate (Section 6.3).

Relative to the US evidence, the paper moves the question to the intensive margin, where the penalty operates through how much mothers who hold jobs can work rather than through whether they hold them; it measures exposure with an index calibrated to realized Israeli remote work rather than imported unchanged; and it treats the small number of occupation clusters as part of the evidence rather than as a nuisance. To our knowledge it also provides the first estimate of the Israeli motherhood penalty in hours conditional on working. Section 2 reviews the literature, Sections 3 and 4 describe the data and the raw patterns, Section 5 sets out the design, Section 6 reports the results, and Sections 7–9 interpret them and state their limits.

2 Literature Review

The motherhood penalty and the flexibility mechanism. The child penalty is one of the most robust facts in labor economics. Danish administrative data show that women’s earnings fall by roughly 20% in the long run relative to men’s after a first birth, through reduced hours, exit from the labor market and moves into family-friendly, lower-paying jobs (Kleven, Landaïs and Søgaaard, 2019), and the same pattern appears in all six countries of a cross-country comparison, with the size of the penalty tracking gender norms rather than family policy (Kleven et al., 2019). Hours are one of the channels, which is why an intervention that changes the time cost of combining work and care could move the intensive margin even where it does not move employment. The remaining gap has been located in the structure of jobs (Goldin, 2014): firms in many occupations pay disproportionately for long, contiguous, on-site hours, and workers who cannot supply them, disproportionately mothers, pay a penalty out of proportion to the hours they forgo. Remote work lowers the cost of supplying a given number of hours, by removing the commute, letting hours be split around care and permitting a virtual presence in demanding professions. Both premises have empirical support: workers, and women in particular, give up wages for the option to work from home (Mas and Pallais, 2017), and the skilled gender gap is concentrated where long hours are rewarded and narrows where substitutes for household time are cheap (Cortés and Pan, 2019). Remote work may also cut the other way, through lower visibility and slower advancement, so the sign is an empirical question. Goldin’s framework says where to look for it: in the gradient of the response across occupations by teleworkability, not in its average. For Israel, the sorting of women into lower-paying firms, even within the same industry, is a first-order driver of the gender wage gap, while pay differences within firms contribute little (Buzaglo-Baris, 2023), a reminder that any occupation-level measure captures only part of the heterogeneity.

What remote work does to hours and to gender gaps. Work from home settled at roughly a quarter of paid US workdays after 2021 (Barrero, Bloom and Davis, 2021, 2023), freeing time that workers report splitting between paid work, care and leisure (Aksoy et al., 2023). The direct evidence on hours points both ways: information-technology professionals who moved home in 2020 worked more hours while output fell slightly, so that measured productivity dropped by 8–19% (Gibbs, Mengel and Siemroth, 2023), while American time diaries show that on remote days part of the time saved goes to family time and, for women, to household production, and that teleworking mothers report more interruptions to their workday (Pabilonia and Vernon, 2022). Those two results bracket the question this paper asks of mothers: whether the time remote work frees is returned to paid hours or absorbed by care. Among call-center workers, who selects into remote

work differs sharply from what remote work does to them (Emanuel and Harrington, 2024), which is why the design below measures *exposure* from occupational content rather than from realized remote-work status. The experimental evidence on gender gaps is encouraging: randomly assigned home working raised both minutes worked per shift and output (Bloom et al., 2015), and randomly assigned flexibility shifted the division of household tasks toward men (Angelici and Profeta, 2024). The acute 2020 evidence is not: women in WFH-feasible jobs absorbed a larger share of the childcare that school closures created (Adams-Prassl et al., 2020), as anticipated early in the pandemic (Alon et al., 2020), and in Israel the 2020 lockdown reallocated paid and unpaid labor within households in gender-unequal ways (Yaish, Mandel and Kristal, 2021). This paper uses post-acute data (2021–2023) rather than data from the initial shock. The paper closest to ours is the US study (Harrington and Kahn, 2025) whose employment and income results the Introduction summarizes; their employment effect is driven by fields with high returns to hours and inflexible demands on time, and their income result is an intensive-margin finding of which hours are one channel, so our hours estimate is its closest analogue, without treating the two countries or the two outcomes as equivalent. For Israel, estimated motherhood penalties in employment and earnings differ across ethno-religious groups (Budig, Kraus and Levanon, 2023); there is, to our knowledge, no published Israeli estimate of the penalty in hours conditional on working, so the level we report has no direct domestic benchmark.

Measurement and identification. Occupations have been classified by whether their tasks can be performed entirely at home, a criterion 37% of US jobs meet (Dingel and Neiman, 2020); that classification, mapped to two-digit ISCO-08 codes, is the starting point for our exposure index, calibrated against realized Israeli behavior (Section 3.2), for which a descriptive baseline exists (Madhala and Bental, 2021). Because the third difference is a continuous exposure score rather than a treated/untreated split, the design belongs to the continuous-treatment difference-in-differences family (Callaway, Goodman-Bacon and Sant’Anna, 2024), two of whose results on how to read a single slope coefficient are taken up in Section 5.

3 Data and Institutional Context

3.1 Source and sample

The data are individual-level microdata from the Israeli Central Bureau of Statistics Labor Force Survey for the survey years 2017–2019 and 2021–2023. The analysis sample is women aged 25–59 (CBS age-group codes 3–7). A woman is a *mother* if she has any child under 17 in the household and childless otherwise; *Post* equals one for survey

years 2021 and later. The comparison group therefore includes women whose children are grown and women who become mothers within the window (Section 8). The year 2020 is excluded because its arrangements were emergency improvisation rather than the settled remote-work practice whose effect we want to measure, and because no 2020 CBS extract exists for this project.

The primary outcome is *usual weekly work hours*, recorded by the CBS in bins and assigned each bin’s median (Appendix A gives the coding and a 2017 recording correction). Hours are the only intensive-margin outcome the survey supports: the extract records how a wage is paid but not its amount. Hours are defined for women who are employed *and* worked in the survey’s reference week, in every year, a harmonization made necessary by a different zero-hours recording convention in the 2017 extract (Appendix A); it makes the hours population identical across years at the cost of about 10% of each year’s employed sample and narrows the estimand to usual hours among women who were working (Section 8). The secondary outcome is a binary employment indicator (unemployed and not-in-labor-force are both zero). All regressions include the same categorical controls: marital status, religion, age group, district of residence and highest educational attainment (six groups); age enters as the CBS age-group code because the extract contains no continuous age. CBS survey weights are deliberately not applied in any outcome regression (Section 8). Replication code is listed in Appendix E.

3.2 Constructing WFH exposure

Realized WFH status is itself a labor-market outcome, jointly determined with hours, so the mechanism test needs an occupation-level measure of *exposure* to remote work that the individual does not choose. We build three, each reported.

External. The Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) at the two-digit ISCO-08 level. This is pre-period by construction, but it is a US task-based measure and misclassifies occupations where Israeli practice diverged: teaching scores 0.966 on it, yet Israeli schools were fully in person throughout the 2022–23 survey years (Appendix B).

Calibrated (primary). The rule that selects the primary measure is fixed ex ante: the score the headline uses must be anchored on pre-period task content, so that it cannot be built from the post-2021 hours it is meant to explain, and must not place in the treated tail occupations that Israeli institutions kept in person. The external index fails the second condition and the realized index the first; the calibrated index is the only one of the three that meets both, and the other two are reported as bounds in Section 6.3. It replaces the external score with the realized 2022–23 Israeli WFH share only where the two are far apart and the gap is statistically distinguishable from noise; ten of the forty occupations are swapped, teaching and clerical support among them (rule, scores and fit

in Appendix B). Because the swap test draws on 2022–23 data, inside the post-period, the calibrated measure is *partly post-treatment*, a documented compromise rather than a fully pre-treatment index; Section 6.3 tests whether the calibration is what produces the result.

Realized. The realized Israeli WFH share by occupation, anchored to 2021. Post-treatment by construction, it is reported as a bound.

A fourth measure serves three roles. For the employment DDD, occupation is undefined for the non-employed, so that specification uses a pre-period (2017–2019) shift-share index by demographic cell (sex, age group, education, district, marital status, religion and continent of birth; 8,884 cells), equal to the mean of the calibrated occupation score over the cell’s pre-period occupational composition and defined for employed and non-employed women alike. The same index stratifies the selection correction of Section 5.4, and because a woman’s post-2021 occupation cannot move it, it also bounds occupational sorting in Section 6.3. Its dispersion is a third of the occupation-level index’s, so coefficients on the two are compared per standard deviation, never per unit.

4 Descriptive Statistics

4.1 The sample in summary

Table 1 describes the sample on either side of the treatment/control split. Mothers work *fewer* hours than childless women when they work (38.68 versus 40.06) while being slightly *more* likely to be employed (0.78 versus 0.76), so the level of the penalty is visible on the intensive margin; whether it changed after 2021 is not something the table can show. The two groups are imbalanced in ways the controls must absorb: mothers are the better-educated group (48% with an academic degree against 40%), overwhelmingly married (85% against 47%), and about 0.44 of an age-group code younger, an imbalance that tracks WFH exposure (Appendix Table A3).

4.2 The raw gap and its exposure gradient

Mothers work 38.54 hours per week before 2021 and 38.84 after, a gain of 0.31 hours; childless women move from 40.04 to 40.08. The difference between those two changes, 0.27 hours, is the difference-in-differences of equation (1) without a single control, and the regression DiD of Table 2, 0.228 (SE 0.181), is within four hundredths of an hour of it. Mothers remain well below childless women in both periods, and the raw gap moves by less than a third of an hour across the three pre-period years (Appendix Figure A1), so the groups were not diverging before treatment.

Figure 1 computes the same difference-in-differences separately within each quartile of occupational WFH exposure. The bottom three quartiles return -0.32 , 0.27 and -0.02

Table 1: Summary statistics: women aged 25–59, by mother status

	Childless women	Mothers	Difference (SE)
<i>Panel A: outcomes</i>			
Observations	133,646	239,095	
Employment rate	0.7574	0.7828	0.0254 (0.0031)
Usual weekly hours (ref.-week workers), mean	40.06	38.68	−1.38 (0.09)
standard deviation	[12.12]	[11.62]	
observations with usual hours	93,982	164,193	
<i>Panel B: characteristics</i>			
Age-group code (CBS 3–7), mean	5.40	4.96	−0.44 (0.01)
Arab (%)	17.73	17.78	0.05 (0.33)
<i>Marital status (%)</i>			
Married	46.55	84.57	38.02 (0.35)
Married, living separately	3.58	0.84	−2.74 (0.11)
Divorced	13.13	7.70	−5.42 (0.24)
Widowed	2.75	0.99	−1.76 (0.11)
Single, never married	34.00	5.90	−28.10 (0.31)
<i>Religion of household head (%)</i>			
Jewish	73.08	79.50	6.42 (0.36)
Christian	7.12	2.23	−4.89 (0.18)
Muslim	13.78	14.89	1.11 (0.30)
Druze	1.88	1.64	−0.24 (0.12)
Other	4.15	1.75	−2.40 (0.14)
<i>District of residence (%)</i>			
Jerusalem	8.76	12.31	3.56 (0.26)
North	17.52	16.58	−0.94 (0.32)
Haifa	14.38	11.92	−2.45 (0.28)
Center	23.99	26.52	2.53 (0.36)
Tel Aviv	21.55	14.90	−6.65 (0.33)
South	11.66	12.37	0.72 (0.27)
Judea and Samaria	2.15	5.39	3.24 (0.15)
<i>Education (%)</i>			
Below high school	11.37	6.63	−4.74 (0.23)
High school, no matriculation	12.70	11.95	−0.74 (0.24)
Matriculation (Bagrut)	22.73	18.82	−3.90 (0.29)
Post-secondary, non-academic	11.50	13.68	2.18 (0.24)
Academic degree (BA/MA/PhD)	40.24	47.97	7.73 (0.39)
Other / no certificate	1.47	0.95	−0.52 (0.07)

Israeli CBS Labor Force Survey, survey years 2017–2019 and 2021–2023, women aged 25–59; a mother has any child under 17 in the household. Weekly hours are conditional on working in the reference week, with the standard deviation in brackets and their own observation count. The difference column is mothers minus childless women, with a standard error clustered by individual. All figures are *unweighted*. The Arab share is the CBS population-group code used for the subgroup analysis; religion is a separate item recorded for the household head. Marital status, religion, district, age group and education are the five controls in every regression.

hours, none distinguishable from zero; the top quartile returns 1.55 hours (95% CI [0.74, 2.36]). The near-zero average is therefore a weighted average of three nulls and one substantial positive difference. The figure is a description rather than an independent causal test: occupational composition and time-varying confounders can differ across quartiles, which is what the regressions of Section 6 address. The response is not linear in exposure: the bottom three quartiles span exposure values of 0.00 to 0.30 and the top quartile 0.30 to 0.75, so most of the regressor’s dispersion, and the identifying contrast, sits in one tail.

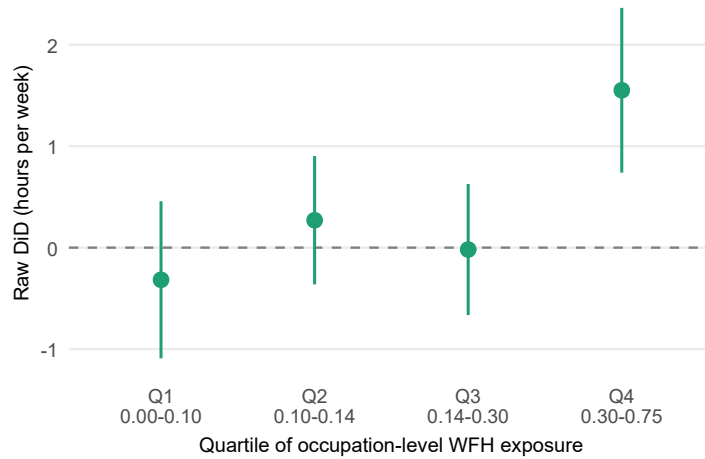


Figure 1: Raw hours difference-in-differences by quartile of WFH exposure

Notes: Women aged 25–59 who worked in the reference week and carry a matched occupation code. Each point is the raw difference-in-differences within one quartile of the occupation-level calibrated exposure measure, from unweighted cell means with no controls; bars are 95% confidence intervals clustered by individual. Breakpoints are computed on pre-period rows and applied to all years; each quartile’s exposure range is printed beneath its tick. The 1.55 hours in the top quartile is that quartile’s own difference-in-differences; the raw contrast between the top and bottom quartiles is $1.55 - (-0.32) = 1.87$ hours.

5 Empirical Strategy

5.1 Difference-in-differences

For outcome Y_{it} (weekly hours conditional on working that week, or employment) we estimate

$$Y_{it} = \beta_0 + \beta_1 \text{Mother}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Mother}_i \times \text{Post}_t) + X'_{it} \gamma + \varepsilon_{it}, \quad (1)$$

where X_{it} is the control vector of Section 3 and β_3 is the differential change in the outcome for mothers relative to childless women after the shift to remote work.

5.2 Triple differences

The flexibility framework (Goldin, 2014) yields one testable hypothesis: after 2021, mothers' usual hours conditional on working rise relative to childless women's in proportion to their occupation's WFH exposure, with no response where the work cannot be done from home. The exposure gradient, not the average change, is the object of interest, so we interact the DiD with occupational WFH exposure:

$$\begin{aligned} Y_{it} = & \alpha + \delta_1 \text{Mother}_i + \delta_2 \text{Post}_t + \delta_3 \text{WFH_Exposure}_{j(i)} + \delta_4 (\text{Mother}_i \times \text{Post}_t) \\ & + \delta_5 (\text{Mother}_i \times \text{WFH_Exposure}_{j(i)}) + \delta_6 (\text{Post}_t \times \text{WFH_Exposure}_{j(i)}) \quad (2) \\ & + \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)}) + X'_{it} \gamma + \varepsilon_{it}, \end{aligned}$$

where $j(i)$ is individual i 's occupation. δ_7 asks whether the post-shift change for mothers relative to non-mothers is larger in more exposed occupations; δ_6 absorbs any post-2021 change in teleworkable occupations common to mothers and childless women, δ_5 the pre-existing level of the penalty across occupations, and δ_4 is the post-shift change for mothers at *zero* exposure, not an average effect. The DDD requires a single parallel-trends assumption (Olden and Møen, 2022): that the mother/non-mother gap in high-versus low-exposure jobs would have evolved in parallel absent the shift to remote work. Insignificant pre-period coefficients are evidence about that assumption, not a verification of it. For the hours DDD, WFH_Exposure_j is the calibrated occupation-level score and the sample is employed women matched to a two-digit occupation; for the employment DDD, WFH_Exposure is the cell-based index, the sample is all women, and a $\text{Mother} \times \text{age-group}$ interaction is added for the pre-period age imbalance.

Equation (2) makes each lower-order occupation-level difference linear in exposure, an unnecessary restriction with forty occupations, and comparing effects *across* exposure levels, which is what a single slope does, requires the stronger condition that occupations at different exposure levels would have followed the same path under any common level of exposure, and the linear slope is a weighted average of such comparisons whose weights need not be positive (Callaway, Goodman-Bacon and Sant'Anna, 2024). We therefore estimate two relaxations alongside it. The *saturated* specification replaces every lower-order term with fixed effects,

$$Y_{it} = \delta_7 (\text{Mother}_i \times \text{Post}_t \times \text{WFH_Exposure}_{j(i)}) + X'_{it} \gamma + \lambda_{j(i),t} + \mu_{j(i),\text{Mother}_i} + \nu_{\text{Mother}_i,t} + \varepsilon_{it}, \quad (3)$$

where λ_{jt} , $\mu_{j,m}$ and $\nu_{m,t}$ are occupation-by-year, occupation-by-mother and mother-by-year effects, which absorb every lower-order term of equation (2) and any occupation-specific year shock. The *binned* specification replaces $\text{WFH_Exposure}_{j(i)}$ with indicators for the pre-period exposure quartile the occupation falls in, the bins of Figure 1, so that the top-versus-bottom contrast carries a standard error. For each DDD we also report

the minimum detectable effect on the reference distribution the model’s inference uses, $\text{MDE} = \text{SE}(t_{0.975} + t_{0.80})$, which is $2.87 \times \text{SE}$ with 39 degrees of freedom and $2.80 \times \text{SE}$ under the normal, so that a null can be read as “genuinely null” or “underpowered.”

5.3 Inference

The survey is a rotating panel in which most respondents appear about four times, so equation (1) and every descriptive interval are clustered by individual (Appendix A gives the panel counts). For the hours DDD the regressor is constant within a two-digit occupation, and standard errors are clustered at that level, leaving forty clusters; the reported p -values use a t reference distribution with $G - 1 = 39$ degrees of freedom and a cluster-count adjustment to the variance. Forty is few enough that we do not rest on that approximation: for the headline coefficient and for the pre-period coefficients of the triple-difference event study we also report wild cluster bootstrap p -values, with Rademacher weights, the null imposed and 9,999 draws (Roodman et al., 2019; Fischer and Roodman, 2021), and Section 5.5 adds a permutation test that relies on no asymptotic approximation at all. The three procedures answer related but different questions, and where they disagree the disagreement is part of the evidence. The joint pre-period test remains the analytic Wald F , since a joint bootstrap test is not available in the implementation used. The employment DDD is clustered on the age-by-education-by-district cell (about 210 clusters).

5.4 Selection into employment: Lee bounds

Hours are observed only for women who are working, and working is itself an outcome of the same treatment. We therefore report trimming bounds (Lee, 2009): the share of post-period mothers observed with hours in excess of a parallel-trends counterfactual is trimmed from the top of that cell’s hours distribution for a lower bound and from the bottom for an upper bound, the model is refit, and the Imbens–Manski interval for the identified set (Imbens and Manski, 2004) is reported. For the DDD the trimming is done within quartiles of the cell-based index, which is defined for the non-employed. Appendix C gives the construction and its assumptions. The bounds address excess selection under monotonicity; they cannot address selection that violates it, nor differences in absence unrelated to excess selection.

5.5 Diagnostics, falsification, and the fathers’ comparison

We estimate event-study versions of equation (1) with Mother $\times$ year interactions (reference year 2019) for both outcomes and test the pre-period coefficients jointly; because those test the difference-in-differences assumption rather than the triple-differences one, we also estimate the event-study version of equation (2) itself, with Post replaced by

survey-year interactions at every level, and apply the same joint test to its pre-period coefficients.

The falsification test is a permutation test on the exposure score. Under the sharp null that teleworkability is unrelated to the post-2021 change in the motherhood gap, the forty occupation scores are exchangeable, so we reassign them at random across the forty occupation codes 999 times, holding every woman’s occupation and the mother/period structure fixed, refit equation (2), and record the cluster-robust t -statistic of the triple interaction; the permutation p -value is the share of draws, counting the observed one, whose $|t|$ reaches the observed value (Phipson and Smyth, 2010). Because occupation is observed when the outcome is, we also check whether remote work drew mothers into teleworkable occupations after 2021, with a difference-in-differences that takes the exposure score itself as the outcome, and bound the channel by re-estimating the hours DDD on the pre-period cell index of Section 3.2, which a woman’s later occupation cannot move. The remaining checks, collected in Table 3, re-estimate the hours DDD with the exposure measure, the age control, the sample, the outcome coding and the inference procedure varied in turn, and re-estimate it by the age of the youngest child, separately for Jewish and Arab women, and on the male subsample with “mother” read as “father.” The last is a comparison, not a placebo: fathers in teleworkable occupations are exposed to remote work exactly as mothers are, so a negative gradient would be evidence on household reallocation rather than on a confound. Coefficients across disjoint subsamples are compared with two-sample z -tests.

6 Results

Three findings organize this section. The average motherhood penalty in weekly hours did not change after the shift to remote work. It did narrow where the work can be done from home: the post-2021 change in the mother/childless gap rises with an occupation’s WFH exposure, with the response concentrated in the most teleworkable quartile of occupations. That gradient survives the design’s checks on selection, sorting, calibration and specification, is largest for mothers of young children and reversed in sign for fathers, while the employment margin is too imprecisely estimated to support a comparable conclusion. Tables 2–5 and Figure 2 carry the estimates; the text reports the pattern. Significance is denoted *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, · $p < 0.1$, with standard errors in parentheses.

6.1 No average change in the hours penalty

Column (1) of Table 2 estimates equation (1). Mothers work 1.739 fewer hours per week than childless women, and that penalty did not measurably change after 2021: Mother $\times$

Post is 0.228 (SE 0.181), and with a minimum detectable effect of about 0.51 hours the estimate rules out average gains much above half an hour a week. It is a reasonably precise null. Panel (a) of Figure 2 supports the comparison: the pre-period coefficients are indistinguishable from zero and pass a joint Wald test ($F(2, 63, 198) = 0.949$, $p = 0.387$), and the post-period path is flat apart from a larger 2023 coefficient. Survey-year effects in place of the single Post dummy, and the selection bounds of Appendix C, leave the reading unchanged. Nothing that follows rests on this average.

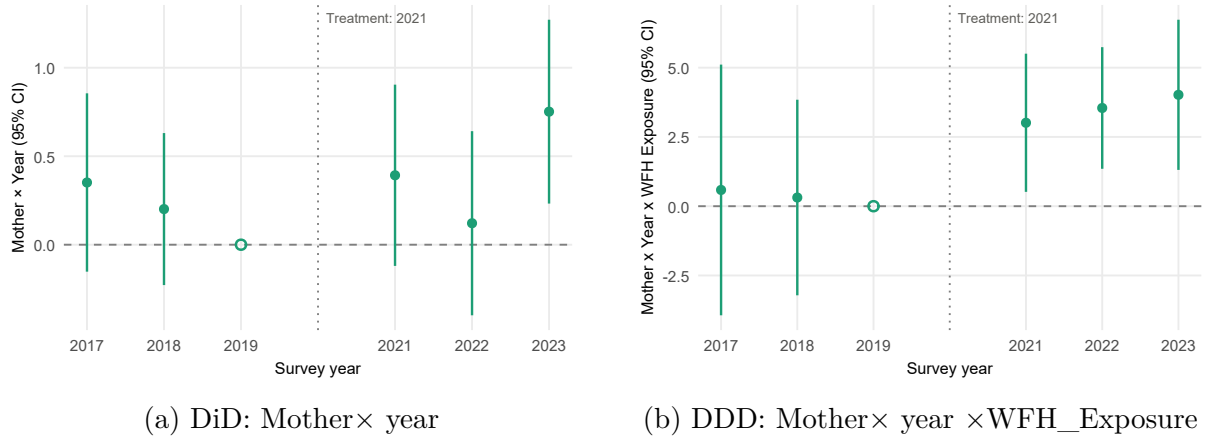


Figure 2: Event studies for weekly hours, reference year 2019

Notes: Vertical bars are 95% confidence intervals; the hollow point at 2019 is the omitted reference year, and the gap between 2019 and 2021 is the excluded 2020 survey year. Panel (a): women working in the reference week ($N = 251,857$), standard controls, standard errors clustered by individual. Coefficients (SE): 2017, 0.352 (0.257); 2018, 0.201 (0.219); 2021, 0.392 (0.261); 2022, 0.121 (0.265); 2023, 0.752** (0.265). Joint test on 2017–2018: $F(2, 63, 198) = 0.949$, $p = 0.387$. Panel (b): sample, controls, exposure and clustering as in column (2) of Table 2 ($N = 246,326$, 40 clusters), with the year main effects and both two-way year interactions alongside the plotted triple interaction. Coefficients (SE), hours per unit of exposure: 2017, 0.587 (2.237); 2018, 0.313 (1.745); 2021, 3.011* (1.233); 2022, 3.545** (1.085); 2023, 4.020** (1.340). Joint test on 2017–2018: $F(2, 39) = 0.041$, $p = 0.960$; wild cluster bootstrap p -values (Rademacher, 9,999 draws) 0.838 for 2017, 0.884 for 2018 and 0.852 for their sum.

6.2 Hours rose where the work can be done from home

Column (2) of Table 2 interacts the DiD with the calibrated occupation-level exposure index. The triple interaction is 3.224** (SE 1.022): among working mothers, those in more WFH-exposed occupations saw a larger post-2021 increase in weekly hours relative to childless women in the same occupations. That is 0.64 hours per standard deviation of the index (0.199), and 1.71 hours between the mean exposures of the bottom and top quartiles of Figure 1, the pattern that figure shows in the raw cell means. The implied change at zero exposure is -0.539 (SE 0.336), so mothers in jobs that cannot be done from home gained nothing, and the gradient is net of whatever happened to hours in

teleworkable occupations for all women. What changed after 2021 is not mothers' hours in general but mothers' hours where remote work was possible.

Columns (3) and (4) relax the functional form and tell the same story. With the three sets of fixed effects of equation (3) absorbing every lower-order term, the triple interaction is 2.876 (SE 0.950), a tenth below the pooled estimate and still significant at the 1% level. With quartile indicators in place of the score, the post-2021 change in the gap relative to the least teleworkable quartile is positive but insignificant in the second and third quartiles and 1.844 hours (SE 0.595, $p < 0.01$) in the top one: a step at the top of the exposure distribution more than a smooth gradient. That quartile holds six of the forty occupations, but the pooled and saturated columns, which use all forty, agree with it, and no single occupation drives it (Section 6.4).

Panel (b) of Figure 2 is the event-study version of equation (2), the pre-trend test the triple difference itself requires (Section 5.5). The pre-period coefficients are near zero and jointly insignificant ($F(2, 39) = 0.041$, $p = 0.960$; wild cluster bootstrap $p = 0.85$ on their sum), and the post-period path steps up in 2021 and rises through 2023, every year straddling the pooled estimate. The test has limited power: with forty clusters the pre-period standard errors are of the same order as the effect, so each pre-period interval contains values as large as the estimate itself. It shows no sign of a violation; it does not rule one out.

How precisely the gradient is estimated depends on the inference procedure, and we report three. The analytic standard error, clustered by occupation on a t_{39} reference, puts the estimate 3.15 standard errors from zero ($p = 0.003$). The two procedures built for forty clusters are less emphatic. The permutation test gives $p = 0.032$: 31 of 999 random reassignments of the forty scores across occupations produce a $|t|$ at least as large as the observed one (Appendix Figure A2), so a design with this cluster structure produces a gradient this large about three times in a hundred when exposure carries no information. The wild cluster bootstrap gives $p = 0.056$, with a 95% confidence set of $[-0.26, 5.84]$. We report all three rather than choosing among them.

6.3 Selection, sorting, and the calibrated regressor

Three features of the design could produce an exposure gradient with no change in mothers' labor supply: selection into work, sorting of mothers across occupations, and an index calibrated on post-period data. Table 3 reports the tests.

Selection into work. Hours are observed only for women who work, and working is itself an outcome. Trimming the excess selection among post-period mothers within each quartile of the cell-based index (Appendix C) barely moves the estimate in either direction, and the Imbens–Manski 95% interval for the identified set, $[1.021, 5.324]$, excludes zero.

Table 2: Intensive-margin DiD and DDD: weekly hours, reference-week workers

	(1) DiD	(2) DDD	(3) DDD, saturated	(4) DDD, quartiles
Mother $\times$ Post $\times$ WFH_Exposure		3.224** (1.022)	2.876** (0.950)	
Mother $\times$ Post $\times$ Q2 (vs. Q1)				0.753 (0.483)
Mother $\times$ Post $\times$ Q3 (vs. Q1)				0.600 (0.485)
Mother $\times$ Post $\times$ Q4 (vs. Q1)				1.844** (0.595)
Mother $\times$ Post	0.228 (0.181)	-0.539 (0.336)		-0.472 (0.346)
Mother $\times$ WFH_Exposure		-1.295 (1.053)		
Post $\times$ WFH_Exposure		-0.548 (0.810)		
WFH_Exposure		1.128 (3.912)		
Mother	-1.739*** (0.144)	-1.449*** (0.351)		-1.527** (0.452)
Post	-0.057 (0.148)	0.053 (0.273)		-0.130 (0.367)
Constant	32.355*** (0.318)	32.115*** (1.540)		31.581*** (2.051)
MDE for triple interaction (t_{39})		2.938		
SD of exposure, estimation sample		0.199		
Triple interaction per SD of exposure		0.640		
Occupations in Q1 / Q2 / Q3 / Q4				16 / 7 / 11 / 6
Fixed effects	None	None	Three sets	None
Observations	251,857	246,326	246,324	246,326
R^2	0.040	0.041	0.135	0.048
Clustering	individual	occupation (40)	occupation (40)	occupation (40)

Dependent variable: usual weekly work hours, women working in the reference week; standard controls in every column. Columns (2)–(4) use the calibrated occupation-level exposure index on column (1)’s sample less the rows with no matched two-digit occupation. Column (3) is equation (3), whose three sets of fixed effects (occupation $\times$ year, occupation $\times$ mother, mother $\times$ year) absorb every lower-order term. Column (4) replaces the score with indicators for its pre-period quartile (Figure 1’s bins); its lower-order quartile terms are estimated but not shown, and the number of occupations per bin is listed because occupation is the cluster. Columns (2)–(4) use a t reference distribution with 39 degrees of freedom (Section 5.3), which the MDE also uses ($\alpha = 0.05$, power 0.80, multiplier 2.87); the wild-bootstrap and permutation p -values for column (2)’s triple interaction are in Table 3. Column (2)’s Mother $\times$ Post is the effect at *zero* exposure, not an average effect. Coefficients and standard errors to three decimals throughout. No coefficient is dropped for collinearity in any column.

Sorting across occupations. If remote work drew mothers into teleworkable occupations after 2021, the occupation score would itself respond, and it does not: as a dependent variable it gives a Mother $\times$ Post under one hundredth of its standard deviation, the probability of being in the top quartile moves by 0.4 percentage points (SE 0.7), and the event-study version is flat in every year. The hours DDD on the pre-period demographic-cell index, which a woman’s later occupation cannot move, is positive but imprecise (0.139 hours per standard deviation of that index, SE 0.159); that index is a far coarser measure of a woman’s own teleworkability, so the sorting DiD is the informative test.

Calibration. The calibrated index draws on realized 2022–23 WFH shares, so the regressor could be absorbing post-period outcomes. The external Dingel–Neiman index (Dingel and Neiman, 2020), the only measure built entirely from pre-treatment information, gives 0.672 (SE 0.884), positive but insignificant and the weakest of the three exposure measures; the realized 2021-anchored index gives a larger, noisier estimate. Three checks hold the sample at all forty occupations. Calibrating on men only, so that no woman in the regression contributes to her own regressor, gives 3.869*** (SE 0.979). Keeping the external score but letting the ten occupations the calibration reclassifies carry their own Mother $\times$ Post change restores a gradient in the external score at two thirds of the headline ($p = 0.09$): the occupations the external index places in the treated tail did not respond. Dropping those ten, so that the two indices coincide on the thirty that remain, gives 4.309** (SE 1.192). The calibration does not produce the result, but it cannot be made into a pre-treatment measure, and a reader who weights the external index above the others will read the headline as directionally supported rather than established.

6.4 Robustness of the gradient

The remaining checks in Table 3 vary the age control, the sample and the outcome coding, and one of them matters. Mothers and childless women differ in age, education and marital status in ways that themselves vary with exposure (Appendix Table A3), and such a gap is what the triple interaction is identified off. A Mother $\times$ age-group interaction leaves the estimate essentially unchanged, but reweighting to balance pre-period age groups attenuates it by about 19% to 2.597* (SE 1.050), still significant at the 5% level but below the design’s minimum detectable effect: the largest single move any check produces.

Everything else stays close to the headline. Excluding the 2023 survey year, whose final quarter includes the start of the October 2023 war (Section 8), gives 2.979* (SE 1.170), while the plain hours DiD stays a null. Removing each of the forty occupations in turn (Appendix Figure A3) leaves the estimate between 2.415 and 3.580, within one standard error of the headline and significant at the 5% level in every refit ($p \leq 0.017$). Across every specification in the table that keeps a calibrated index, the triple interaction

lies between 2.4 and 4.3 hours.

Nor does the outcome coding drive it. Hours are a bin-median approximation and the effect is smaller than one bin width, but dropping the two imputed irregular-hours codes changes little, and because 35 and 40 hours are exact bin edges the same specification can be estimated as a linear probability model on either threshold: the probability of usually working at least 35 hours, and at least 40, rises by about 2 percentage points per standard deviation of exposure (1.9 and 2.1) for mothers relative to childless women after 2021. The effect is a shift of mothers in teleworkable jobs across the full-time boundary.

Table 3: Hours DDD: sensitivity of Mother $\times$ Post $\times$ WFH_Exposure

Specification	Coefficient	SE	p	N	Clusters
<i>Exposure measure</i>					
Calibrated (primary)	3.224**	1.022	0.0031	246,326	40
External (Dingel–Neiman)	0.672	0.884	0.4517	246,326	40
Realized (2021 anchor)	5.195*	2.363	0.0346	246,037	36
Calibrated on men only (12 of 40 swapped)	3.869***	0.979	0.0003	246,326	40
External index with swapped-occupation terms (40 occ.)	2.339	1.352	0.0915	246,326	40
Mother $\times$ Post $\times$ Swapped (same model)	−1.347	0.926	0.1537	246,326	40
<i>Age balance</i>					
Age-interacted (+ Mother $\times$ age group)	3.205**	1.055	0.0042	246,326	40
Reweight (pre-period age-group raking)	2.597*	1.050	0.0178	242,552	40
<i>Sample</i>					
Unswapped occupations only	4.309**	1.192	0.0011	132,046	30
Excluding survey year 2023	2.979*	1.170	0.0151	208,833	39
Leave-one-occupation-out (40 refits): range	[2.415, 3.580]		[0.0007, 0.0167]		39
<i>Occupational sorting</i>					
Pre-period cell exposure as regressor, per SD	0.139	0.159	0.3823	247,931	245
Exposure score as outcome: Mother $\times$ Post	0.001	0.003	0.6996	246,326	individual
Top-quartile indicator as outcome: Mother $\times$ Post	0.004	0.007	0.5433	246,326	individual
<i>Outcome coding</i>					
Irregular-hours codes dropped, not imputed	3.056**	1.059	0.0064	232,785	40
Full-time indicator (≥ 35 hours)	0.096**	0.029	0.0020	246,326	40
Long-hours indicator (≥ 40 hours)	0.107**	0.038	0.0075	246,326	40
<i>Inference on the primary estimate</i>					
Two-way clustering (individual, occupation)	3.224**	1.022	0.0031	246,326	40 $\times$ ind.
Wild cluster bootstrap ($B = 9,999$)	3.224	1.022	0.0558	246,326	40
Permutation test (999 reassignments)	3.224		0.0320	246,326	40

Unless a row says otherwise: the triple interaction Mother $\times$ Post $\times$ WFH_Exposure from equation (2), women working in the reference week, standard controls, calibrated exposure, clustered by occupation with the cluster count in the last column, and p the analytic $t(G - 1)$ p -value. The men-only row calibrates the index on men’s realized 2022–23 WFH shares; the swapped-occupation rows keep the external index as regressor on all forty occupations and add a Swapped indicator for the ten calibrated occupations with the full Mother $\times$ Post structure, showing both triple interactions from that one model; the unswapped-occupations row drops those ten. The leave-one-occupation-out row gives the range of the forty refits with one occupation removed (Appendix Figure A3) and the range of their p -values. Under *Occupational sorting*, the first row is the hours DDD with the pre-period demographic-cell index of Section 3.2 as regressor, reported per standard deviation of that index and clustered on the demographic cell; the next two are differences-in-differences with the occupation-level score, and an indicator for its top quartile, as the dependent variable, clustered by individual. The two indicator outcomes under *Outcome coding* are linear probability models on bin-crossing outcomes. The wild-bootstrap row reports the analytic coefficient and standard error with the bootstrap p -value; the permutation row reports the permutation p -value, which has no standard error.

6.5 Who responded: child age, population group, and fathers

If remote work relaxes a time constraint that binds through child care, the response should be largest for mothers of young children and smallest for mothers of teenagers, and it is. Table 4 re-estimates the DiD and DDD for mothers whose youngest child is aged 0–4, 5–9, 10–14 or 15–17, each against the same childless women. The average effect is a null in every group, and the exposure gradient declines monotonically with the child’s age, from 3.656** hours per unit of exposure (SE 1.079) for mothers of under-fives to 1.862 (SE 0.935) for mothers of 15–17-year-olds, half as large. The four regressions share a control group, so their intervals overlap and are not independent; the pattern is a gradient consistent with the mechanism rather than a set of statistically separated effects, and the estimate is not carried by mothers of teenagers.

Table 5 splits the sample by population group and re-estimates the models on men. Jewish women reproduce the pooled pattern. Arab women’s gradient is nominally twice as large but statistically indistinguishable from it ($z = 0.942$, $p = 0.35$) on a subsample nine times smaller: imprecise rather than a documented heterogeneous effect.

For men, with “mother” read as “father,” the gradient has the opposite sign, -1.762 (SE 1.015, $p = 0.09$), and is distinguishable from the mothers’ ($z = 3.461$, $p = 0.0005$). Fathers in teleworkable occupations are exposed to remote work exactly as mothers are (Section 5.5), so the comparison is not a placebo and a null was never the prediction. What it shows is that a macro trend common to all workers in teleworkable jobs cannot produce a positive gradient for mothers beside a negative one for fathers, and that opposite gradients are what within-household reallocation would look like if mothers’ additional paid hours were matched by fathers’ additional care hours. We report that as a reading the data are consistent with, not a result the design identifies: mothers and fathers are not linked in the data, and the fathers’ coefficient is itself imprecise.

Table 4: Hours DiD and DDD by age of the youngest child

Sample of mothers	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure		
	Coef.	SE	N	Coef.	SE	N
All mothers (Table 2)	0.228	0.181	251,857	3.224**	1.022	246,326
Youngest child aged 0–4	0.127	0.211	158,424	3.656**	1.079	154,499
Youngest child aged 5–9	0.162	0.253	130,782	3.211*	1.311	127,419
Youngest child aged 10–14	0.159	0.278	123,280	2.741	1.525	120,084
Youngest child aged 15–17	0.561	0.348	107,061	1.862	0.935	104,079

Weekly hours, reference-week workers, standard controls; each row compares the named group of mothers with all childless women, so the rows are not independent samples. DiD clustered by individual; DDD clustered by occupation, calibrated exposure. The first row repeats Table 2. No coefficient is dropped for collinearity in any column.

Table 5: Hours DiD and DDD by subgroup, and the fathers’ comparison

Subgroup	DiD: Mother $\times$ Post			DDD: Mother $\times$ Post $\times$ WFH_Exposure		
	Coef.	SE	N	Coef.	SE	N
All women (primary)	0.228	0.181	251,857	3.224**	1.022	246,326
Jewish women	0.302	0.203	214,449	2.847**	0.913	209,727
Arab women	−0.072	0.479	24,975	5.507*	2.674	24,307
Men (fathers vs. childless)	−0.400*	0.188	261,543	−1.762*	1.015	251,779
95% CI, Jewish	[−0.095, 0.699]			[1.058, 4.635]		
95% CI, Arab	[−1.010, 0.866]			[0.266, 10.748]		
Jewish vs. Arab, z (p)	0.720 (0.471)			−0.942 (0.346)		
Women vs. men, z (p)	2.406 (0.016)			3.461 (0.001)		

Weekly hours, reference-week workers, standard controls. DiD clustered by individual; DDD clustered by occupation. Men: “Mother” read as “Father,” a comparison population rather than a placebo (Section 5.5). Two-sample z -tests as described there. Confidence intervals use the normal critical value. The Arab-women DiD drops the religion category ‘other’, constant on that sample; the Arab-women DDD drops the religion category ‘other’, constant on that sample; no other column drops a coefficient.

6.6 The extensive margin

Appendix Table A2 reports equation (1) with employment as the outcome. The DiD is -0.0053 (SE 0.0052), an informative null with a standard error of half a percentage point whose pre-period coefficients pass their joint test. The employment DDD with the cell-based index is a null of a different kind: the design could detect a triple interaction no smaller than 1.43 percentage points per standard deviation of the cell index, and against the US estimate of 0.78 percentage points per 10% rise in WFH (Harrington and Kahn, 2025) it is underpowered by a factor of two to three. The shortfall is structural, a binary outcome and a regressor with a standard deviation of 0.07, and re-clustering, combining the two margins and refining the cell partition did not close it. The employment DDD is therefore *uninformative*, evidence neither for nor against a WFH-mediated employment effect, and the two nulls should not be read as one verdict.

7 Discussion

Among mothers who remain in work there is no post-2021 hours effect on average, and a large one in teleworkable occupations: a triple interaction of about 3.2 hours per unit of exposure and an implied change at zero exposure that is if anything negative. This is consistent with WFH access relaxing a binding time constraint for mothers in teleworkable jobs, the flexibility mechanism (Goldin, 2014), and hard to reconcile with a general recovery common to all women, which would have raised hours across occupations rather than only where the work can be done from home. What makes the finding credible is the pattern of evidence rather than any one p -value: a saturated specification that changes

nothing, a top-quartile contrast significant on its own, a leave-one-out range that never leaves the headline’s standard error, a monotone child-age gradient, an opposite-signed fathers’ gradient and a selection correction that barely moves the bounds. The gradient is identified off a concrete contrast: ICT professionals stayed high on every exposure measure, while teaching and clerical support, which Israeli practice kept in person, moved sharply down.

The magnitude is economically meaningful where it applies. For a mother at the top quartile’s mean exposure the regression implies a post-2021 change of about 1.4 hours a week ($-0.54 + 3.22 \times 0.59$), and the raw top-quartile difference-in-differences of Figure 1 is 1.55. That 1.4 is the fitted change *at* top-quartile exposure, distinct from the 1.71-hour fitted contrast *between* the quartile means. The penalty it is set against must be the penalty in those same jobs, not the average one: column (2) of Table 2 puts the pre-period motherhood gap at $-1.45 - 1.30 \times 0.59 \approx -2.2$ hours at that exposure, so mothers in the most teleworkable jobs recovered roughly three fifths of their penalty after 2021. At the bottom quartile’s mean exposure of 0.06 the implied change is -0.35 hours: mothers in the least teleworkable jobs recovered none of it. The size matches the commuting arithmetic: the time saved is about 72 minutes per day worked from home (Aksoy et al., 2023), so two remote days a week free roughly two and a half hours, and a mother who returns most of that time to paid work is a mother whose hours rise by about the amount estimated here. The timing is consistent with a multi-year adjustment: the DDD event study rises from 3.0 in 2021 to 4.0 in 2023 (on the 2023 year, see Section 8).

A general recovery is not the only alternative to that mechanism; three demand-side accounts also fit a gradient by teleworkability. Employers in professional and ICT occupations may have changed the contracts they offer mothers after 2021, with no change in mothers’ constraints; the saturated specification absorbs any occupation-by-year shock common to mothers and childless women, so this needs a change aimed at mothers in teleworkable occupations specifically. The 2021–22 technology hiring boom raised hours in exposed occupations for everyone; the fathers’ opposite-signed gradient rules out a shock common to all workers in those occupations, but not one that fell on mothers alone. And usual hours are self-reported: if mothers working from home report the hours they are available while fathers report the hours they work, the two gradients would diverge as observed with no change in labor supplied; the child-age gradient weighs against this, since a reporting difference has no reason to be twice as large for mothers of infants as for mothers of teenagers. None of the three is excluded by the design, and we read the exposure gradient as the time-constraint mechanism’s signature rather than its proof.

Why hours and not employment? The employment DDD is not a failed test of the mechanism but an uninformative one (Section 6.6). Substantively the two results are coherent: the effect operates on *how much* mothers in teleworkable jobs work, not on *whether* they are employed, which is a leading channel of the child penalty (Kleven, Landaís and Sør-

gaard, 2019), and it is where our findings depart from the US evidence on employment and income (Harrington and Kahn, 2025). The policy reading is correspondingly narrow: WFH availability matters for how intensively mothers who already hold *teleworkable* jobs can work, not for whether mothers in general hold jobs; and access to those jobs is itself unequally distributed across mothers, so the population-wide gain is smaller than the gradient alone suggests and cannot be sized without representative weighting.

8 Limitations

Identification and exposure. The triple difference’s own pre-trend test (Section 6.3) shows no visible violation but lacks the power to rule one out. The exposure measure that delivers the headline draws on post-period Israeli data for its calibration, and although the checks of Section 6.3 show the calibration does not produce the result, the fully pre-treatment external index yields a positive and insignificant estimate. The sorting check finds no post-2021 movement of mothers across exposure levels but bounds the effect only imprecisely on a regressor fixed before treatment. Exposure is assigned at the two-digit occupation level, forty groups, whereas sorting across firms, even within the same industry, is a first-order driver of Israeli gender gaps (Buzaglo-Baris, 2023); within-occupation heterogeneity in teleworkability is classical measurement error in the regressor and attenuates the triple interaction toward zero, and the calibrated score mixes two scales (Appendix B). Occupation codes masked by the CBS for disclosure control are dropped when building exposure; a proxy check finds no strong evidence of bias from this. How precisely the headline is estimated is the range of the three inference procedures in Section 6.2, not the analytic p -value. The permutation test itself assumes that the forty exposure scores are exchangeable across occupations under its null.

The narrowed hours estimand. Hours are usual hours among women who were working, not among all women holding a job, and roughly 10% of each year’s employed sample is excluded on that basis. The exclusion is not innocuous for the triple interaction because absence is related to exposure: it is highest in the least teleworkable quartile, and among mothers it fell after 2021 in the top quartile while rising in the bottom one (Appendix A). The bounds of Section 5.4 therefore define selection on the observed-hours sample; they handle only *excess* selection under monotonicity and found little to correct, which is not the same as the concern being absent.

Sample and period. The 2023 coefficient is the largest in both event studies, and the 2023 survey year includes the quarter in which the war that began on 7 October 2023 disrupted the labor market: reserve call-ups, which bear on the fathers’ comparison, and school closures and evacuations, which bear on mothers’ hours. The estimate survives

dropping the year and 2021 and 2022 are individually significant on the triple difference, but the 2023 point should be held more loosely. Early 2021 likewise included a third lockdown, so the post-period is “post-acute” more cleanly in its later years than in its first. The control group is any woman aged 25–59 with no child under 17 in the household, so it includes women whose children are grown and women who become mothers within the window, as in the US study (Harrington and Kahn, 2025); if part of it responds to the treatment as mothers do, the DiD is biased toward zero. Age enters as a categorical code because the extract has no continuous age, and mothers and non-mothers are imbalanced on it in a way that tracks exposure (Appendix Table A3); the age-interacted and reweighted specifications of Section 6.4 address this and the estimate survives both.

Survey weights. CBS survey weights are intentionally not applied in any outcome regression; reported rates and coefficients are unweighted estimates on the analysis sample, not population-representative statistics. The one exception is internal to constructing the cell-based exposure index.

9 Conclusion

Using Israeli survey data, we ask whether the expansion of remote work coincided with a change in the motherhood penalty in usual weekly hours. The average change among women observed working is small, whereas the post-2021 change in the mother/comparison gap increases with an occupation’s calibrated WFH exposure. The strongest contrast is at the top of the exposure distribution and among mothers with younger children. Those patterns are consistent with greater scheduling flexibility helping some already employed mothers supply more paid hours.

The result is conditional and does not establish a universal benefit from remote work. The exposure score partly uses post-period WFH information; the wholly external score yields a weaker result, pre-trend tests have limited power, and inference varies with the procedure. Employment is an important separate margin, but its exposure interaction is too imprecise here to settle the question. The data also contain no earnings amount. Stronger evidence would link hours with earnings and household time use, measure exposure before treatment in the Israeli setting, and estimate women’s and men’s responses together on a design that better separates access to WFH from occupation choice.

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A Data and sample details

Hours coding. The CBS records usual weekly hours in bins; each bin is assigned its median (4, 11, 18, 25.5, 32, 37, 42, 47, 54.5 and 78.5 hours) and the two irregular-hours codes are imputed from observed medians within the same period. Mid-distribution bins are about five hours wide, so an effect smaller than one bin width, such as the headline estimate, is best read as a shift in the share of women crossing a bin boundary. In the 2017 file, employed respondents absent from work in the reference week had their usual hours recorded as zero, whereas from 2018 onward the same respondents were assigned a real value; left uncorrected, this enters the pre-period as roughly 5,200 spurious zeros, and because absence is more common among mothers (12.4% of employed mothers against 7.3% of childless women) they fall unevenly across the comparison the paper makes. Defining hours on the reference-week workers in *every* year costs about 10% of each year’s employed sample, makes the hours population identical across years, and narrows the estimand to usual hours among women who were working.

Reference-week absence and exposure. Absence is related to exposure. Among employed women with a matched occupation, by the quartiles of Figure 1, the absence shares are 14.4%, 8.5%, 9.5% and 9.6%. Among mothers the top-quartile share fell from 12.1% to 10.4% after 2021 and the bottom-quartile share rose from 15.5% to 16.9%; among childless women the top-quartile share is 6.1% in both periods. Movement across the absence margin is therefore part of what selection into the observed-hours sample means, which is why the bounds of Appendix C define selection on that sample.

Panel structure. The survey is a rotating panel. The 258,175 reference-week workers of Table 1 come from 64,602 distinct respondents, about four observations each and two to three within a single survey year, with 96% of rows belonging to someone who appears more than once; the regressions use 251,857 of those rows from 63,199 respondents, the difference being the 6,318 rows whose education code is missing (education is the only control with missing values). Treating the observations as independent would understate standard errors by a factor of roughly 1.8, which is why every DiD and every descriptive

interval is clustered by individual. The panel does not support within-person identification: of the 80,560 women in the full sample, 1.9% are interviewed on both sides of the pre/post boundary, so an individual fixed-effects design was considered and rejected.

Hours year by year. Panel (a) of Figure A1 unpools the pre-period. The raw mother-minus-non-mother gap is -1.37 hours in 2017, -1.48 in 2018 and -1.67 in 2019, three years within a third of an hour of each other and each within the others' confidence bounds: the groups were not diverging before treatment. The post-period path is -1.37 , -1.53 and -0.80 , not monotone and not individually distinguishable, so the timing reading in the Discussion rests on the event studies of Figure 2, not on this series.

Commuting. Mothers were the *more* mobile group in every year, working outside their locality of residence 48.2% of the time in 2019 against 46.1% of childless women, and their commuting fell less in 2021 (1.5 against 3.9 percentage points, a difference of 2.39 points, SE 1.17). This is consistent with remote work relaxing a time constraint for women who were already commuting; it is descriptive evidence only, and no specification uses locality of work as an outcome.

B Exposure construction

The external score. The Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) reaches the two-digit ISCO-08 level through the BLS crosswalk between ISCO-08 and the 2010 SOC (Bureau of Labor Statistics, 2012). Each O*NET-SOC occupation in their published classification is matched to every ISCO-08 unit group the crosswalk lists for its six-digit SOC code, and the score of a sub-major group is the unweighted mean of their binary indicator over the matched pairs; armed-forces groups have no counterpart in their data and carry no score. The replication code (Appendix E) re-derives the forty scores (Table A1) from the two public source files and checks them against the file the pipeline reads. Teaching (ISCO 23) is the clearest case of a US task-based score misclassifying Israeli practice: it scores 0.966 on the external index, yet Israeli schools, closed and reopened in stages through 2020 and early 2021 (Somekh et al., 2021; Hale et al., 2021), were fully in person throughout the 2022–23 survey years, and teachers did not work from home.

The calibration rule. The external score is replaced by the realized 2022–23 Israeli WFH share only where the gap between the two is large (greater than 0.5) *and* distinguishable from sampling noise by a one-sided test at the 95% level with a standard error clustered by individual. Ten of the forty two-digit occupations are swapped: teaching falls from 0.966 to 0.069, clerical support (ISCO 41) from 1.000 to 0.098, ICT professionals

(ISCO 25) from 1.000 to 0.360. The realized shares are computed on men and women together; Table 3 re-derives them from men alone. The realized index used as a bound is the realized Israeli WFH share by occupation, anchored to 2021 (no 2020 extract exists) with a minimum of 200 observations per occupation.

Fit. Across the forty occupations, on men and women pooled over 2021–23, the calibrated score has a correlation of 0.53 with the share who worked from home in the reference week and an occupation-size-weighted slope of 0.40 (SE 0.09), so it is relevant. The raw external index fits better in levels (correlation 0.74), because swapping ten occupations onto realized shares while leaving thirty on task-based probabilities mixes two scales. The calibration’s contribution is not a better cross-sectional fit but the removal of ten grossly mis-scored occupations from the treated tail. Table A1 lists, for each occupation, the external score, the realized 2022–23 share the calibration tests it against, whether the score was swapped, the calibrated score the hours DDD uses, and the realized reference-week share over 2021–23; panel (b) of Figure A1 plots the last against the calibrated score.

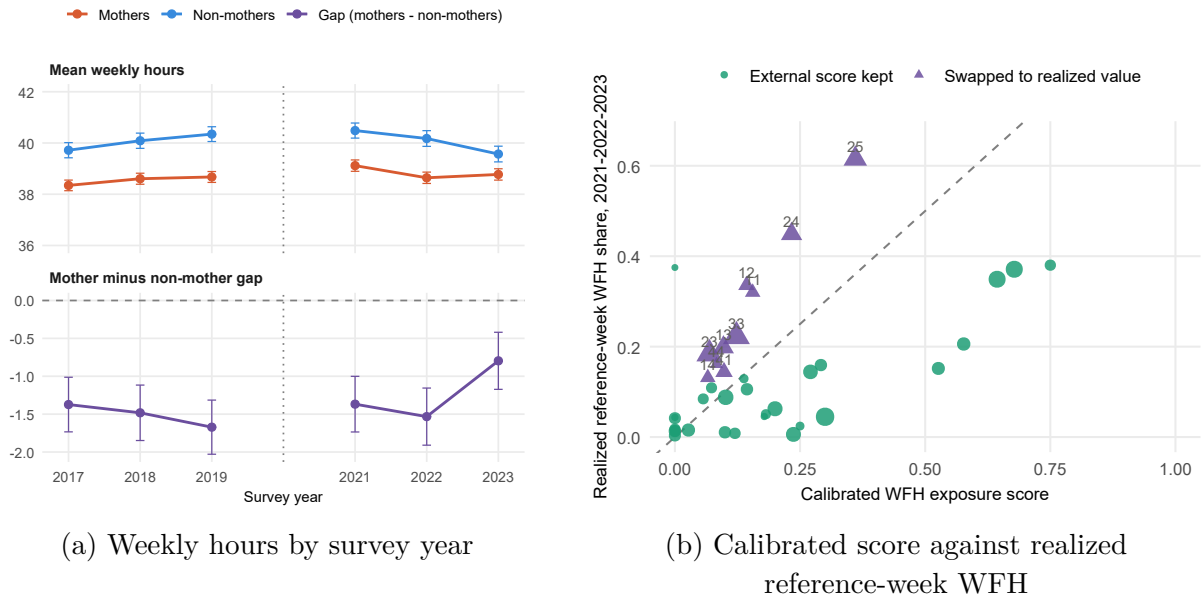


Figure A1: Weekly hours by survey year, and the exposure first stage
Notes: Panel (a): women aged 25–59 who worked in the reference week, unweighted means with no controls; the upper panel plots each group’s mean weekly hours on an axis spanning 36 to 42 hours, the lower panel their difference, with 95% confidence intervals clustered by individual; the dotted rule marks the excluded 2020 survey year, and the series break there. Panel (b): one point per two-digit occupation, sized by the pooled employed sample in 2021–23; triangles are the ten occupations the calibration swapped to their realized value, labelled by ISCO-08 code; the dashed line is the 45-degree line. The swapped occupations sit above it because their calibrated value is the usual-place share, which is lower than the reference-week share everywhere; the thirty unswapped ones lie below it, the visual form of the scale-mixing described above.

Table A1: External, realized and calibrated WFH exposure by two-digit occupation

ISCO	Occupation (ISCO-08 sub-major group)	Ext.	Real. 22–23	Swap	Calib.	Ref.-wk 21–23	<i>N</i>
11	Chief executives and legislators	0.900	0.155	yes	0.155	0.320	2,855
12	Administrative and commercial managers	0.892	0.143	yes	0.143	0.336	3,859
13	Production and specialised services managers	0.638	0.097	yes	0.097	0.199	9,990
14	Hospitality, retail and other services managers	0.773	0.066	yes	0.066	0.132	2,827
21	Science and engineering professionals	0.644	0.210		0.644	0.349	13,228
22	Health professionals	0.101	0.059		0.101	0.088	9,665
23	Teaching professionals	0.966	0.069	yes	0.069	0.184	20,496
24	Business and administration professionals	0.943	0.233	yes	0.233	0.449	10,479
25	ICT professionals	1.000	0.360	yes	0.360	0.615	14,638
26	Legal, social and cultural professionals	0.678	0.215		0.678	0.371	13,344
31	Science and engineering associate professionals	0.144	0.059		0.144	0.106	3,926
32	Health associate professionals	0.057	0.080		0.057	0.084	2,203
33	Business and administration associate professionals	0.684	0.123	yes	0.123	0.223	22,415
34	Legal, social and cultural associate professionals	0.577	0.142		0.577	0.206	5,480
35	ICT technicians	0.750	0.244		0.750	0.380	2,491
41	General and keyboard clerks	1.000	0.098	yes	0.098	0.144	4,846
42	Customer services clerks	0.292	0.096		0.292	0.159	3,875
43	Numerical and material recording clerks	0.526	0.071		0.526	0.152	4,948
44	Other clerical support workers	0.625	0.083	yes	0.083	0.161	1,049
51	Personal service workers	0.271	0.146		0.271	0.144	7,808
52	Sales workers	0.200	0.050		0.200	0.063	8,704
53	Personal care workers	0.300	0.046		0.300	0.045	17,206
54	Protective services workers	0.100	0.008		0.100	0.011	3,969
61	Market-oriented skilled agricultural workers	0.182	0.027		0.182	0.051	1,325
62	Market-oriented skilled forestry, fishery and hunting workers	0.179	0.000		0.179	0.045	22
63	Subsistence farmers, fishers, hunters and gatherers	0.000	0.375		0.000	0.375	8
71	Building and related trades workers	0.027	0.008		0.027	0.016	5,208
72	Metal, machinery and related trades workers	0.000	0.008		0.000	0.013	3,982
73	Handicraft and printing workers	0.138	0.118		0.138	0.129	666
74	Electrical and electronic trades workers	0.000	0.022		0.000	0.042	3,424
75	Food, wood, garment and other craft workers	0.073	0.122		0.073	0.109	2,304
81	Stationary plant and machine operators	0.000	0.019		0.000	0.017	2,407
82	Assemblers	0.000	0.012		0.000	0.015	1,166
83	Drivers and mobile plant operators	0.237	0.006		0.237	0.006	8,480
91	Cleaners and helpers	0.000	0.004		0.000	0.003	3,474
92	Agricultural, forestry and fishery labourers	0.000	0.007		0.000	0.016	192
93	Labourers in mining, construction, manufacturing and transport	0.120	0.006		0.120	0.008	2,659
94	Food preparation assistants	0.000	0.013		0.000	0.011	649
95	Street and related sales and service workers	0.000	0.000		0.000	0.043	23
96	Refuse workers and other elementary workers	0.250	0.022		0.250	0.024	493

Ext.: the Dingel–Neiman teleworkability score (Dingel and Neiman, 2020) at the two-digit ISCO-08 level, via the crosswalk described above. Real. 22–23: share of employed respondents (men and women, all ages in the extract) who usually work from home in 2022–23, the calibration’s benchmark; Swap: the calibration replaced the external score with it. Calib.: the score the hours DDD uses. Ref.-wk 21–23: share who worked from home in the survey reference week over 2021–23, men and women pooled, with *N* the pooled employed sample carrying that item. Codes 1–3 (armed forces) carry no external score and are not in the sample.

C The selection correction

Let $s_{ab} = \Pr(\text{hours observed} \mid \text{Mother} = a, \text{Post} = b)$, the probability of being employed and working in the reference week. Under parallel trends in selection the counterfactual for post-period mothers is $s_{11}^* = s_{10} + s_{01} - s_{00}$. If s_{11} exceeds it, the excess share $1 - s_{11}^*/s_{11}$ is trimmed from the ($\text{Mother} = 1, \text{Post} = 1$) hours distribution, from the top for the lower bound and from the bottom for the upper bound, and the model is re-estimated on each trimmed sample. Trimming one tail assumes monotone selection, and the construction handles only excess selection in that cell. For the DDD the counterfactual and trim are computed within each quartile of the cell-based exposure index, which assumes selection depends on exposure only through that quartile, and equation (2) is refit on the re-combined sample with the occupation-level regressor and occupation clustering. Because the parameter is partially identified, the Imbens–Manski interval for the identified set is reported with the bounds (Imbens and Manski, 2004).

For the DiD, $s_{11} = 0.696$ against $s_{11}^* = 0.683$, and trimming the 1.9% excess gives bounds of -0.517 (SE 0.178) and 0.802 (SE 0.179) around the 0.228 point estimate, with Imbens–Manski 95% interval $[-0.810, 1.096]$. For the DDD the excess is present in all four quartiles but modest, a trim of 0.80% to 4.48%, 1,396 of 72,820 post-period working mothers; the bounds are 3.126 (SE 1.144) and 3.443 (SE 1.022) around an untrimmed 3.232 (SE 1.054), which differs from Table 2’s 3.224 because this sample additionally requires a matched cell-based exposure, and the Imbens–Manski interval is $[1.021, 5.324]$.

D Additional results

Figure A2 plots the reference distribution of the permutation test: under the sharp null the central 95% of cluster-robust t -statistics lie between -2.96 and 2.60 , and the observed 3.15 is exceeded in absolute value by 31 of 999 draws. Table A2 reports the extensive-margin DiD and DDD discussed in Section 6.6. Table A3 reports the pre-period covariate balance between mothers and childless women within each exposure quartile of the hours DDD’s estimation sample, and Figure A3 the forty leave-one-occupation-out refits of the headline estimate.

E Replication code

The analysis is a set of R scripts. `main.R` orchestrates the pipeline: it loads and cleans the CBS microdata (cached beside the raw files), builds the exposure measures, fits every model, and writes the results to `outputs/` and the tables to `paper/tables/`; `Rscript main.R` reproduces every number in this paper. Each file under `scripts/` defines one function; the multi-function robustness chains live under `robustness/`. The listings below

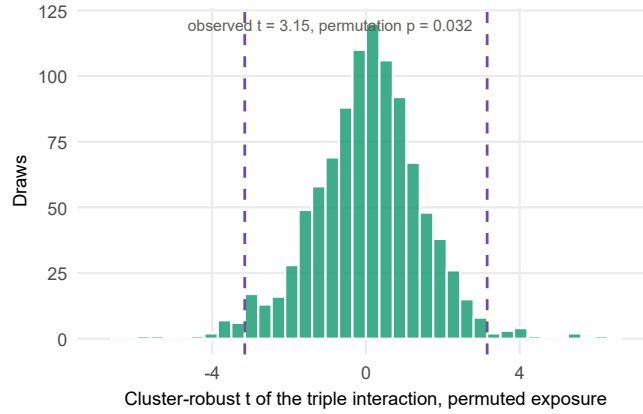


Figure A2: Permutation distribution of the hours DDD triple interaction

Notes: Histogram of the cluster-robust t -statistic of Mother $\times$ Post $\times$ WFH_Exposure over 999 random reassignments of the forty exposure scores across occupations, with the same sample, controls and clustering as column (2) of Table 2. Dashed lines mark $\pm$ the observed statistic (3.15); $p = 0.032$ counts the observed draw (Phipson and Smyth, 2010).

Table A2: Extensive margin: employment DiD and DDD

	(1) DiD	(2) DDD, cell-based exposure
Mother $\times$ Post $\times$ WFH_Exposure		-0.0257 (0.0725)
Mother $\times$ Post	-0.0053 (0.0052)	0.0065 (0.0168)
Mother $\times$ WFH_Exposure		-0.1404 (0.0803)
Post $\times$ WFH_Exposure		-0.1555** (0.0566)
WFH_Exposure		0.0131 (0.0565)
Mother	-0.0034 (0.0041)	
Post	0.0153*** (0.0043)	
Constant	0.4926*** (0.0085)	
MDE ($\alpha = 0.05$, power 0.80), per unit		0.2031
MDE per SD of exposure		0.0143
as % of baseline employment rate		1.85%
Observations	364,784	355,984
R^2	0.227	0.221
Clustering	individual	demographic cell (≈ 210)

Dependent variable: employment indicator; full sample of employed and non-employed women; standard controls in both columns, four decimals because the outcome is a probability. Column (2) adds Mother $\times$ age-group interactions, so its Mother and Post main effects are not separately reported, and uses the demographic-cell index of Section 3.2, whose standard deviation on this sample is 0.0706; the per-SD MDE row is the interpretable one. Baseline employment rate 0.7737. No coefficient is dropped for collinearity in any column.



Figure A3: Leave-one-occupation-out: the hours DDD triple interaction with each occupation removed

Notes: Each point is $\text{Mother} \times \text{Post} \times \text{WFH_Exposure}$ from equation (2) re-estimated without the occupation named on the axis (column (2) of Table 2 on 39 occupations), with its 95% confidence interval clustered by occupation; rows are sorted by the estimate. The dashed line is the headline estimate on all forty occupations and the shaded band one headline standard error either side of it; the dotted line marks zero. Triangles would mark refits outside the band; there are none.

Table A3: Pre-period covariate balance by quartile of occupation-level WFH exposure

Mothers minus childless women, pre-period	Q1	Q2	Q3	Q4
Age-group code (CBS 3–7), mean	−0.54*** (0.03)	−0.44*** (0.03)	−0.33*** (0.03)	0.08* (0.04)
Married (%)	32.2*** (1.1)	33.7*** (1.1)	41.5*** (0.9)	43.1*** (1.3)
Academic degree (%)	10.8*** (1.2)	4.5*** (1.2)	1.5 (1.0)	1.8 (1.2)
Below high school (%)	−3.2*** (0.6)	−1.7*** (0.4)	−3.3*** (0.5)	−0.2 (0.2)
Jewish household head (%)	1.6 (1.0)	0.5 (0.7)	14.3*** (0.9)	3.1*** (0.8)
Center or Tel Aviv district (%)	−4.4*** (1.2)	−3.4** (1.2)	−8.8*** (1.1)	−4.5** (1.4)
Arab (%)	1.9* (0.9)	0.2 (0.6)	3.7*** (0.6)	−0.9 (0.6)
Mothers	22,976	24,664	22,562	16,377
Childless women	11,420	11,586	17,244	8,352
Occupations	15	7	11	6
Exposure range	0.00–0.10	0.10–0.14	0.14–0.30	0.36–0.75

Women working in the reference week in 2017–2019 with a matched two-digit occupation, the hours DDD’s pre-period estimation sample, split into the exposure quartiles of Figure 1. Each cell is the mother-minus-childless difference in the row variable within the quartile, with a standard error clustered by individual in parentheses; shares are in percentage points, the age-group code in code units (CBS codes 3–7 span ages 25–59 in five-year bands). Stars follow the paper’s legend on a normal reference. Unweighted.

are the estimation code, grouped by role: data construction, the exposure measures, the DiD and DDD specifications, the event studies, the selection bounds, inference, and the calibration and sorting checks. Everything else, the pipeline orchestration, the wrappers and loops that call the primary functions on subsamples, the helpers that draw the figures, assemble the tables and export the results, and the secondary diagnostics, is named in Table A4 with the function it wraps or the place its number appears. Regressions use `fixest`; the wild cluster bootstrap uses `fwildclusterboot`. The raw CBS microdata are not distributed with the code.

Table A4: Scripts not listed: pipeline, wrappers, figures, tables, export and guards

File	Role
<i>Pipeline and wrappers</i>	
<code>main.R</code> (all but section 8b)	Orchestration: loads and caches the data, calls every function below in order, exports the results
<code>scripts/placebo_male_frame.R</code>	Loads, validates and caches the male subsample
<code>scripts/hours_gender_placebo.R</code>	<code>run_intensive_margin_reg()</code> and the hours DDD formula on the male subsample; Table 5
<code>scripts/hours_ddd_by_child_age.R</code>	The two primary functions on each child-age bin's mothers plus all childless women; Table 4
<code>scripts/hours_ddd_leave_one_out.R</code>	Forty refits of <code>run_hours_ddd_regression()</code> , one occupation removed each time; Figure A3
<code>robustness/pretrend_wald_test.R</code>	<code>fixest::wald()</code> on the pre-2020 event-study terms; Figure 2 notes
<code>scripts/tidy_event_study_coefs.R</code>	Extracts the year-interaction coefficients for plotting and export
<code>scripts/occupation_exposure_breaks.R</code> , <code>scripts/assign_wfh_quartile.R</code>	Pre-period worker-weighted quartile edges of the exposure score, and the <code>cut()</code> that applies them
<i>Descriptive figures and tables</i>	
<code>scripts/descriptive_table.R</code>	Table 1
<code>scripts/hours_descriptive_plots.R</code>	Figure A1, panel (a)
<code>scripts/hours_dose_response.R</code>	Figure 1 and the quartile edges it supplies
<code>scripts/wfh_share_by_year.R</code>	Israeli WFH shares by year (Introduction)
<code>scripts/absence_by_exposure_quartile.R</code>	Reference-week absence by quartile (Appendix A)
<code>scripts/wfh_occupation_first_stage.R</code>	Table A1; Figure A1, panel (b)
<code>scripts/build_balance_by_exposure_quartile.R</code>	Table A3
<code>scripts/comparative_statistics.R</code>	Commuting series (Appendix A)
<code>scripts/build_event_study_plot.R</code>	Figure 2
<code>scripts/build_permutation_plot.R</code>	Figure A2
<code>scripts/build_leave_one_out_plot.R</code>	Figure A3
<code>scripts/clustered_se.R</code> , <code>scripts/paper_theme.R</code>	Clustered standard errors for the descriptives; figure theme
<i>Generated tables and export</i>	
<code>scripts/build_paper_tables.R</code>	Every table in this paper, assembled from the fitted models
<code>scripts/export_paper_tables.R</code>	
<code>scripts/tex_coef_cell.R</code> , <code>scripts/format_tex_table_body.R</code>	Cell and tabular formatting for the above
<code>scripts/export_results.R</code> , <code>scripts/export_paper_figures.R</code>	CSV and PNG export of every result; vector PDFs of the paper's figures
<i>Guards and secondary diagnostics</i>	
<code>scripts/validation.R</code>	Data-quality and schema checks on the cleaned data
<code>scripts/Diagnostics.R</code>	Employment 2×2 table, missingness audits and the employment event study (the hours specification with Employed as outcome); Section 6.6
<code>scripts/ddd_mde_diagnostics.R</code>	The minimum detectable effect of Section 5, $SE(t_{0.975} + t_{0.80})$, per unit, SD and IQR of the regressor
<code>scripts/isco_masking_diagnostics.R</code>	Masked-versus-unmasked occupation check within ISCO major groups (Section 8)
<code>scripts/ddd_collinearity_diagnostics.R</code>	Dropped-coefficient guard; collinearity report for the employment DDD
<code>robustness/balance_test.R</code>	Pre-period covariate balance by cell-exposure quartile

Data

scripts/data_processing.R

```
library(tidyverse)
```

```
# DEFAULT_CONTROLS is the single source of truth for the regression controls.
DEFAULT_CONTROLS <- c("MatzavMishpachti", "Dat", "GilNK", "MachozMegurim", "
  ↳ TeudaGvoha")
```

```
load_and_clean_data <- function(folder_path, sex_filter = c("women", "men")) {
```

```
  sex_filter <- match.arg(sex_filter)
  min_code <- if (sex_filter == "women") 2 else 1
```

```
  # 1. Load raw data
  if (!dir.exists(folder_path)) stop("Target data folder not found.")
```

```
  data_raw <- list.files(folder_path, pattern = "\\*.csv$", full.names = TRUE) %>%
    set_names() %>%
    map_df(~read_csv(.x, guess_max = 50000, show_col_types = FALSE), .id = "file_
      ↳ source")
```

```
  # 2. Filter: sex, ages 25-59 (GilNK 3~7), survey years excluding 2020
  filtered_df <- data_raw %>%
    filter(
```

```
      Min == min_code,
      between(GilNK, 3, 7),
      ShnatSeker %in% c(2017, 2018, 2019, 2021, 2022, 2023)
    )
```

```
  # 3. Derived variables
  # ShaotAvodaBederechKlalNK bin -> median usual weekly hours (codebook: bin bounds)
  hour_bin_median <- c('0' = 0, '1' = 4, '2' = 11, '3' = 18, '4' = 25.5, '5' = 32,
    '6' = 37, '7' = 42, '8' = 47, '9' = 54.5, '10' = 78.5)
```

```
  # Values in the 90s of the raw hour items are CBS codes, not hours; as.numeric()
    ↳ first because
  # an empty column reads as logical.
  hours_or_na <- function(x) {
    v <- suppressWarnings(as.numeric(x))
    if_else(!is.na(v) & v >= 1 & v <= 89, v, NA_real_)
  }
```

```
  mutated_df <- filtered_df %>%
    mutate(
      Mother = as.integer(MisparYeladimAd17MB > 0),
      Post = as.integer(ShnatSeker >= 2021),
      # Employed: Muasak == 1; unemployed and not-in-labor-force are both 0.
      Employed = if_else(!is.na(Muasak) & Muasak == 1, 1L, 0L),
      # WFH block, asked from 2021 only. CBS codes 1 = yes, 2 = no, 9 = unknown; 9
        ↳ maps to NA, not 0.
      WFH = case_when(
        ShnatSeker < 2021 ~ NA_real_,
        AvodaMeHaBayit == 1 ~ 1,
        AvodaMeHaBayit == 2 ~ 0,
        .default = NA_real_ # code 9 ("unknown") and genuine missingness
      ),
      # Reference-week WFH is asked only of those who worked that week, so the absent
        ↳ are NA.
      WFH_RefWeek = case_when(
        ShnatSeker < 2021 ~ NA_real_,
        AvadMeHaBayit == 1 ~ 1,
        AvadMeHaBayit == 2 ~ 0,
        .default = NA_real_
      ),
      # Hours worked from home in the reference week; codes 90-97 are excluded.
      .hrs_home = hours_or_na(KamaShaot),
      .hrs_total = hours_or_na(ShaotAvodaLeMaase),
      WFH_Hours = case_when(
        WFH_RefWeek == 0 ~ 0,
        WFH_RefWeek == 1 ~ .hrs_home,
        .default = NA_real_
      ),
      WFH_Share = case_when(
        WFH_RefWeek == 0 ~ 0,
        WFH_RefWeek == 1 & .hrs_total > 0 ~ pmin(.hrs_home / .hrs_total, 1),
```

```
      .default = NA_real_
    ),
    WFH_Arrangement = factor(
      case_when(
        is.na(WFH_Share) ~ NA_character_,
        WFH_Share == 0 ~ "On-site",
        WFH_Share < 0.9 ~ "Hybrid",
        .default = "Fully remote"
      ),
      levels = c("On-site", "Hybrid", "Fully remote")
    ),
```

```
    # CBS disclosure-masks the occupation code ("XX", "7X"); the mask is recorded
      ↳ and the 1-digit
    # major group recovered where possible.
    .isco_chr = as.character(MishlachYad_ISCO_08_2),
    MishlachYad_ISCO_08_2 = suppressWarnings(as.numeric(.isco_chr)),
    ISCO_masked = !is.na(.isco_chr) & is.na(MishlachYad_ISCO_08_2),
    ISCO1 = suppressWarnings(as.numeric(str_sub(.isco_chr, 1, 1))),
```

```
    # Bin median for the regular-hours codes 0-10; WorkHoursCont is completed below,
      ↳ by period.
    .hour_bin_val = unname(hour_bin_median[as.character(ShaotAvodaBederechKlalNK)])
    ↳ ,
```

```
    # Education, grouped into broader categories (raw TeudaGvoha codes; 99 -> NA)
```

```
    TeudaGvoha = factor(
      case_when(
        TeudaGvoha %in% c(0, 1) ~ "Below High School",
        TeudaGvoha == 2 ~ "High School (no matriculation)",
        TeudaGvoha == 3 ~ "Matriculation (Bagrut)",
        TeudaGvoha == 4 ~ "Post-secondary, non-academic",
        TeudaGvoha %in% c(5, 6, 7) ~ "Academic Degree (BA/MA/PhD)",
        TeudaGvoha %in% c(8, 9) ~ "Other/No Certificate",
        .default = NA_character_
      ),
      levels = c("Below High School", "High School (no matriculation)",
        "Matriculation (Bagrut)", "Post-secondary, non-academic",
        "Academic Degree (BA/MA/PhD)", "Other/No Certificate")
    ),
```

```
    # Country of birth by continent; Israel kept separate; code 7 spans continents
      ↳ ("Other"),
    # code 16 is ambiguous (NA).
```

```
    BirthContinent = factor(case_when(
      SemelEretzLeda == 10 ~ "Israel",
      SemelEretzLeda %in% c(1, 6, 11, 14) ~ "Asia",
      SemelEretzLeda %in% c(2, 8, 12, 15) ~ "Africa",
      SemelEretzLeda %in% c(3, 4, 5, 13) ~ "Europe",
      SemelEretzLeda == 9 ~ "North America",
      SemelEretzLeda == 7 ~ "Other",
      .default = NA_character_
    )),
```

```
    # Commutes outside the locality of residence (DargatNayadut 2~7).
```

```
    WorksOutsideLocality = case_when(
      DargatNayadut == 1 ~ 0L,
      DargatNayadut %in% 2:7 ~ 1L,
      .default = NA_integer_
    ) %>%
```

```
    # WorkHoursCont: bin medians for codes 0-10; the irregular-hours codes 11/12 are
```

```
      ↳ imputed from
```

```
    # the median of regular workers in the matching range, separately by period so a
    # period-specific shift is not blended away. Defined only for the employed who
```

```
      ↳ worked in the
```

```
    # reference week (AvadBeshavua == 1): the 2017 file recorded the employed-but-
```

```
      ↳ absent as zero
```

```
    # usual hours while later years gave them a real value, and absence is mother-
```

```
      ↳ skewed, so the
```

```
    # hours population is harmonized across years at the cost of about 10% of each
```

```
      ↳ year's
```

```
    # employed sample.
```

```
    group_by(Post) %>%
```

```
    mutate(
```

```
      WorkHoursCont = case_when(
        Employed != 1 ~ NA_real_,
        AvadBeshavua != 1 ~ NA_real_,
        ShaotAvodaBederechKlalNK == 0 ~ NA_real_,
        ShaotAvodaBederechKlalNK %in% 0:10 ~ .hour_bin_val,
```

```
        ShaotAvodaBederechKlalNK == 11 ~ median(.hour_bin_val[Employed == 1 &
          ↳ ShaotAvodaBederechKlalNK %in% 1:5], na.rm = TRUE),
        ShaotAvodaBederechKlalNK == 12 ~ median(.hour_bin_val[Employed == 1 &
          ↳ ShaotAvodaBederechKlalNK %in% 6:10], na.rm = TRUE),
        .default = NA_real_
      )
    ) %>%
    ungroup() %>%
    select(~.hour_bin_val, ~.hrs_home, ~.hrs_total, ~.isco_chr) %>%
    mutate(
      across(
        c(MatzavMishpachti, Dat, GilNK, MachozMegurim, MisparHorimYechidim),
        as.factor
      )
    )
```

```
  # 4. Drop unused raw columns
```

```
  cols_to_drop <- c(
    "ShnotLimud", "SugBeitSeferAcharon", "AvadBeshavua2", "ChipesChodesh",
    "KamaPachot", "SibaAvadPachot", "MisparShaotNesafot", "ShaotAvodaLeMaase",
    "ChazerLamasik", "KamaShavuotChipes", "ChipusAvodaMeleah",
    "ZminutLeAvodaMechapsim", "SibatEyZminut", "AvadEyPaamBaaretz",
    "SibaHifisikLaavod", "MatayHifisikLaavod", "ChipesBeShanaAchrona",
    "SibaLoChipesAvoda", "ZminutLeAvodaMityaashim", "MimiMekabelSachar",
    "YeladimAd14PratNK", "GilYeledTzairPratNK", "ShaotAvodaLemaaseNK",
    "MeshechChipusAvodaNK", "ShnotLimudNK", "ShayachAvoda",
    "SibaAvadPachot10Chodashim", "LimudiaVavoda", "MityaashimMechipusAvoda",
    "RamatHaskala_ISCED97", "RamatHaskala_ISCED2011", "ShaotOzeretMBeubad",
    "Pratmugbalkashe", "ShnotLimudLeloYeshivotG", "KamaPachotmechushav",
    "SibaAvadPachotmechushav", "AvadEyPaam", "MimiMekabelSacharMechushav",
    "AvadIkari", "AvodaAcheret", "BeeluShaot", "Beizoderech", "Chaverim",
    "ChipesAvodaAcheret", "ChipesShavuot", "ChipesShavuotMityaesh",
    "ChipusAvodaDmeyAvtala", "ChipusAvodaMimachim", "ChipusAvodaShnatHafsa",
    "ChipusAvodaYachalLehatchil30", "ChipusMeleahMityaesh", "ChipusShaot",
    "ChodeshHafsa", "ChodeshHafsaMityaesh", "ChodeshHatchala",
    "DmeyAvtalaMityaesh", "Esek", "HaskalaMatima", "HavtachatHachnasaMityaesh",
    "HifisikMigbala", "HifisikMigbalaMityaesh", "KamaAvodot", "KoachAdam",
    "LehachlifAvoda", "Lehatchil60", "LoChipesMigbala", "ShnatHafsaMityaesh",
    "SibaHifisikLaavodMityaesh", "SofShavua", "SugMachala", "SugTeuna",
    "YachalLehatchil30Mityaesh", "YamimBashavua", "ZmanLaavoda",
    "KamaPachot_Unified",
```

```
  # Explicit names rather than positional ranges: the 2017 file lacks the boundary
```

```
    ↳ columns
```

```
  "EizeChozemechushav", "HaimMemunemechushav", "HaimMenahelmechushav",
  "KamaKufimmechushav", "KamaSchirimmechushav", "LoAvadMigbalemechushav",
  "MaasikSchirimmechushav", "MeshechChipusAvodaMityaeshNK", "
    ↳ MeshechChipusAvodaMuasakNK",
  "MigzarKalkalimechushav", "SacharMechushavmechushav", "SemelMikzoamechushav",
  "SemelMikzoan", "ShaotAvodaBederechKlalikaritNK", "ShaotAvodaLemaaseikaritNK",
  "SugChozemechushav", "SugMachalaPachotmechushav", "SugTeunaPachotmechushav",
  "MigzarTziburAnafi", "TatTaasuka_Zman", "ChodeshKodem", "ChodeshKodemShaa",
  "MinaHaMigbala", "Mismachim", "Modaot", "OfenAcher", "Oved30", "Pniyalmaasik"
  )
```

```
  # Regex pattern matching any column that starts with these prefixes
```

```
  prefix_pattern <- paste0(
    "(",
    paste(c(
      "Kolel", "MisparMugbalim", "Yeshiva", "ChodeshSeker", "ShnatMidgam",
      "ChodeshMidgam", "MisparNefashotMB", "MisparNefashotNesafot",
      "YeladimAd14MBNK", "MisparNefashotM15MB", "MisparBiltiMuasakim",
      "MisparMuasakimMale", "TtchunatAvoda", "Limudim",
      "MigzarChadarimMB", "TzfifutDiyur", "ShayachimKoachAvoda", "YabeshetLeida",
      "VetekNisuinNK", "MaduaLehachlif", "SherutTaasuka", "IsukLifneyShechipes",
      "Needar", "Aliya", "Imut"
    )), collapse = "|"),
    ")"
  )
```

```
  df <- mutated_df %>%
    select(
      ~any_of(cols_to_drop),
      ~matches(prefix_pattern),
      ~Yeladim0_PratiYeladim15_17Prat,
      ~MisparHachlafaYachasKirvaNK
    ) <- (MisparNefashotG11AvodaV2007.MisparPrat),
    ~ChipusAvodaSherutTaasuka:ChipusAvodaOfenAcher),
    ~RamatDat:BituachLeumi)
  )
```

```

return(df)
}

```

Exposure measures

scripts/wfh_exposure_cells.R

```

library(tidyverse)
library(fixest)

# The external (Dingel & Neiman) teleworkability score by two-digit ISCO-08.
build_exposure_isco2 <- function(path = file.path("data", "israeli_cbs_wfh_2digit.csv"
  ↳ " ")) {
  read_csv(path, show_col_types = FALSE) %>%
    transmute(ISCO2 = as.numeric(isco_2digit), tele_ext = wfh_probability_2d) %>%
    filter(!is.na(tele_ext))
}

# Replaces the external score with the realized 2022-23 Israeli WFH share where
  ↳ Israeli practice
# diverges from the US task-based prediction (teaching: 0.966 external, 0.063 realized
  ↳ ). The swap
# requires the gap to exceed gap_threshold by more than the realized share's cluster-
  ↳ robust
# sampling uncertainty (one-sided test at conf_level), so a thin occupation cannot
  ↳ swap on a
# handful of observations.
calibrate_isco_exposure <- function(cleaned_df, exposure_isco2, wfh_col = "WFH",
  ref_year = c(2022, 2023), gap_threshold = 0.5,
  conf_level = 0.95) {

  df <- cleaned_df %>%
    filter(
      ShnatSeker %in% ref_year,
      Employed == 1,
      !is.na(MishlachYad_ISCO_08_2),
      !is.na(.data[[wfh_col]])
    ) %>%
    mutate(ISCO2 = MishlachYad_ISCO_08_2)

  z <- qnorm(conf_level)

  # A cluster-robust SE is undefined with fewer than two IDPUFs or a constant outcome;
    ↳ those
  # occupations keep the external value. Other fixest errors are reported, not
    ↳ swallowed.
  realized <- df %>%
    group_by(ISCO2) %>%
    group_modify(~ {
      n <- nrow(.x)
      p <- mean(.x[[wfh_col]])
      se_clustered <- NA_real_
      se_na_reason <- NA_character_
      if (n_distinct(.x$IDPUF) < 2) {
        se_na_reason <- "fewer than 2 distinct IDPUF"
      } else if (n_distinct(.x[[wfh_col]]) < 2) {
        se_na_reason <- "constant outcome within occupation"
      } else {
        se_clustered <- tryCatch({
          m <- feols(as.formula(paste(wfh_col, "- 1")), data = .x, cluster = -IDPUF,
            ↳ notes = FALSE)
          unname(se(m)["(Intercept)"])
        }, error = function(e) {
          se_na_reason <- paste("unexpected fixest error:", conditionMessage(e))
          NA_real_
        })
      }
      tibble(n = n, realized_wfh = p, se_clustered = se_clustered, se_na_reason = se_
        ↳ na_reason)
    }) %>%
    ungroup()

  undefined <- filter(realized, is.na(se_clustered))
  if (nrow(undefined) > 0) {
    message(sprintf(

```

```

      "calibrate_isco_exposure: %d of %d occupations kept at their theoretical value
        ↳ (no cluster-robust SE could be computed):",
      nrow(undefined), nrow(realized)
    ))
    print(as.data.frame(select(undefined, ISCO2, n, se_na_reason)))
  }

  exposure_isco2 %>%
    left_join(realized, by = "ISCO2") %>%
    mutate(
      gap = abs(realized_wfh - tele_ext),
      margin = gap - z * se_clustered - gap_threshold,
      swap = !is.na(margin) & margin > 0,
      wfh_exposure_calibrated = if_else(swap, realized_wfh, tele_ext)
    ) %>%
    arrange(desc(gap))
}

# Pre-period (2017-2019) shift-share exposure by demographic cell: the calibrated
  ↳ occupation score
# averaged over each cell's occupational composition.
build_exposure_cells <- function(raw_all, exposure_isco2,
  cell_vars = c("Min", "GilNK", "TeudaGvoha", "
    ↳ MachozMegurim")) {
  raw_all %>%
    filter(ShnatSeker %in% 2017:2019, Muasak == 1, !is.na(MishlachYad_ISCO_08_2)) %>%
    mutate(ISCO2 = suppressWarnings(as.numeric(MishlachYad_ISCO_08_2))) %>%
    filter(!is.na(ISCO2)) %>%
    count(across(all_of(cell_vars)), ISCO2, wt = MishkalSofi, name = "w") %>%
    left_join(exposure_isco2 %>% select(ISCO2, tele_ext), by = "ISCO2") %>%
    filter(!is.na(tele_ext)) %>%
    group_by(across(all_of(cell_vars))) %>%
    summarise(
      WFH_Exposure = weighted.mean(tele_ext, w),
      n_cell = sum(w),
      .groups = "drop"
    )
}

```

scripts/wfh_exposure_index.R

```

# wfh_exposure_index.R
# Realized WFH share by occupation in the anchor year (2021): measured after treatment
  ↳ so a bound
# rather than the baseline. Build it from a frame wider than the analysis sample.
library(tidyverse)
source(file.path("scripts", "data_processing.R"))

build_wfh_exposure_index <- function(cleaned_df, isco_col = "MishlachYad_ISCO_08_2",
  wfh_col = "WFH", ref_year = 2021,
  weight_col = NULL, min_n = 0) {

  # Anchor year, employed, known occupation and WFH status.
  df_year <- cleaned_df %>%
    filter(
      ShnatSeker %in% ref_year,
      Employed == 1,
      !is.na(.data[[isco_col]]),
      !is.na(.data[[wfh_col]])
    )

  # Report how many anchor-year rows carry a disclosure-masked occupation code.
  if ("ISCO_masked" %in% names(cleaned_df)) {
    anchor <- filter(cleaned_df, ShnatSeker %in% ref_year, Employed == 1)
    n_masked <- sum(anchor$ISCO_masked, na.rm = TRUE)
    if (n_masked > 0) {
      years_str <- paste(ref_year, collapse = "-")
      message(sprintf(
        "build_wfh_exposure_index: %d of %d employed %s rows (%.1f%%) have a
          ↳ disclosure-masked ISCO code.",
        n_masked, nrow(anchor), years_str, 100 * n_masked / nrow(anchor)
      ))
    }
  }

  idx <- df_year %>%
    group_by(occupation_code = .data[[isco_col]]) %>%

```

```

  summarise(
    wfh_exposure = if (is.null(weight_col)) {
      mean(.data[[wfh_col]])
    } else {
      weighted.mean(.data[[wfh_col]], .data[[weight_col]], na.rm = TRUE)
    },
    n = n(),
    .groups = "drop"
  ) %>%
  arrange(desc(wfh_exposure))

# min_n drops occupations too thin to carry a stable share (200 in main.R).
if (min_n > 0) {
  n_thin <- sum(idx$n < min_n)
  if (n_thin > 0) {
    message(sprintf("build_wfh_exposure_index: dropping %d occupation(s) with n < %
      ↳ d.",
      n_thin, min_n))
  }
  idx <- filter(idx, n >= min_n)
}

idx
}

```

DiD and DDD specifications

scripts/intensive_margin_regression.R

```

# Primary DiD: usual weekly hours on Mother x Post with the standard controls, on the
  ↳ employed.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))

# year_fe replaces the single Post effect with survey-year effects (a robustness row);
  ↳ Mother:Post
# stays the coefficient of interest.
run_intensive_margin_reg <- function(cleaned_df, controls = DEFAULT_CONTROLS, year_fe
  ↳ = FALSE,
  ref_year = 2019) {

  period_term <- if (isTRUE(year_fe)) sprintf("i(ShnatSeker, ref = %d)", ref_year)
    ↳ else "Post"

  rhs <- paste(
    paste("Mother +", period_term, "+ Mother:Post"),
    paste(controls, collapse = " + "),
    sep = " + "
  )

  formula_hours <- as.formula(paste("WorkHoursCont ~", rhs))

  reg_hours <- feols(formula_hours, data = filter(cleaned_df, Employed == 1), cluster
    ↳ = -IDPUF)

  table_hours <- etable(reg_hours,
    headers = c(if (isTRUE(year_fe)) "WorkHoursCont (year
      ↳ effects)" else "WorkHoursCont"),
    digits = 4)

  print(table_hours)
  return(invisible(list(
    table = table_hours,
    models = list(hours = reg_hours)
  )))
}

```

scripts/basic_regression.R

```

# Secondary DiD: employment on Mother x Post with the standard controls.
library(tidyverse)
library(fixest)

```

```
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))
```

```
basic_reg <- function(cleaned_data) {
  controls <- DEFAULT_CONTROLS

  rhs <- paste(
    "Mother + Post + Mother:Post",
    paste(controls, collapse = " + "),
    sep = " + "
  )

  formula_employed <- as.formula(paste("Employed ~", rhs))

  reg_employed <- feols(formula_employed, data = cleaned_data, cluster = ~IDPUF)
  check_for_dropped_coefficients(reg_employed, "basic_reg()"s Employed model")

  table_basic <- etable(reg_employed,
    headers = c("Employed"),
    digits = 4)

  print(table_basic)
  return(invisible(list(
    table = table_basic,
    models = list(employed = reg_employed)
  )))
}
```

main.R, section 8b (employment DDD)

```
# 8b. Secondary DDD (employment): cell-based exposure, full sample, two specifications
message("Running secondary (employment) DDD (cell-based exposure, calibrated, full
  ↳ sample)...")
ddd_df <- cleaned_df %>%
  left_join(exposure_cells, by = exposure_cell_vars)
message(sprintf(
  "Employment DDD join: %d of %d rows unmatched to an exposure cell (WFH_Exposure NA)
  ↳ ."
))
sum(is.na(ddd_df$WFH_Exposure)), nrow(ddd_df)
))

cell_fe_vars <- c("GilNK", "TeudaGvoha", "MachozMegurim")
other_controls <- setdiff(DEFAULT_CONTROLS, cell_fe_vars)

# Clustered on the (GilNK, TeudaGvoha, MachozMegurim) cell
cell_cluster_formula <- as.formula(paste("-", paste(cell_fe_vars, collapse = "")))

ddd_employment_additive <- feols(
  as.formula(paste("Employed ~ Mother * Post * WFH_Exposure + Mother:GilNK +",
    paste(DEFAULT_CONTROLS, collapse = " + "))),
  data = ddd_df, cluster = cell_cluster_formula
)
ddd_employment_fe <- feols(
  as.formula(paste("Employed ~ Mother * Post * WFH_Exposure + Mother:GilNK +",
    paste(other_controls, collapse = " + ")),
    "|", paste(cell_fe_vars, collapse = ""))),
  data = ddd_df, cluster = cell_cluster_formula
)
check_for_dropped_coefficients(ddd_employment_additive, "employment DDD Spec 1 (
  ↳ additive controls)")
check_for_dropped_coefficients(ddd_employment_fe, "employment DDD Spec 2 (interacted
  ↳ cell FE)")
employment_ddd_table <- etable(
  ddd_employment_additive, ddd_employment_fe,
  headers = c("Spec 1: additive controls", "Spec 2: interacted cell FE"), digits = 4
)
print(employment_ddd_table)

message("Checking Spec 1's collinearity at runtime (see comment above)...")
check_spec1_collinearity(ddd_df, cell_fe_vars, DEFAULT_CONTROLS) # prints its own
  ↳ report
```

scripts/hours_ddd_regression.R

```
# hours_ddd_regression.R
```

```
# Primary triple difference: usual weekly hours on Mother x Post x WFH_Exposure, with
  ↳ the
# occupation-level exposure score joined by two-digit ISCO code, on the employed.
  ↳ Conditioning on
# employment is intrinsic to an hours outcome; the selection it introduces is bounded
  ↳ separately
# by run_hours_ddd_lee_bounds(). Also fits a per-occupation second stage (a repo
  ↳ diagnostic).
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))
source(file.path("scripts", "intensive_margin_regression.R"))

# outcome allows the same specification on an alternative coding of hours; run_
  ↳ mechanism switches
# off the per-occupation second stage.
run_hours_ddd_regression <- function(cleaned_df, exposure_index, controls = DEFAULT_
  ↳ CONTROLS,
                                outcome = "WorkHoursCont",
                                run_mechanism = identical(outcome, "
  ↳ WorkHoursCont")) {
  if (!outcome %in% names(cleaned_df)) {
    stop("run_hours_ddd_regression: outcome column '", outcome, "' is not in cleaned_
  ↳ df.")
  }

  n_employed <- sum(cleaned_df$Employed == 1, na.rm = TRUE)

  # Join each employed row's occupation-level exposure; rows without a matched code
  ↳ drop out.
df_ddd <- cleaned_df %>%
  filter(Employed == 1) %>%
  inner_join(
    exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
    ↳ = wfh_exposure),
    by = "MishlachYad_ISCO_08_2"
  )

  n_matched <- nrow(df_ddd)
  # Occupation is the cluster; the count is returned for the robustness table.
  n_clusters <- n_distinct(df_ddd$MishlachYad_ISCO_08_2)
  message(sprintf(
    "run_hours_ddd_regression: %d of %d employed rows (%.1f%%) retained an occupation
    ↳ -level WFH_Exposure match (dropped: disclosure-masked or unmapped ISCO
    ↳ codes); %d occupation clusters.",
    n_matched, n_employed, 100 * n_matched / n_employed, n_clusters
  ))

  rhs_ddd <- paste(
    "Mother*Post*WFH_Exposure",
    paste(controls, collapse = " + "),
    sep = " + "
  )

  formula_ddd <- as.formula(paste(outcome, "-", rhs_ddd))
  # Clustered on occupation, the level the regressor varies at.
  reg_ddd <- feols(formula_ddd, data = df_ddd, cluster = ~MishlachYad_ISCO_08_2)
  check_for_dropped_coefficients(reg_ddd, "run_hours_ddd_regression()"s triple
    ↳ interaction")

  table_ddd <- etable(reg_ddd, headers = c(paste0(outcome, " (hours DDD)")), digits =
    ↳ 4)
  print(table_ddd)

  if (!isTRUE(run_mechanism)) {
    return(invisible(list(
      table = table_ddd,
      model = reg_ddd,
      n_employed = n_employed,
      n_matched = n_matched,
      n_clusters = n_clusters,
      exposure_vector = df_ddd$WFH_Exposure,
      models = list(ddd = reg_ddd, mechanism = NULL),
      mechanism_data = NULL,
      dropped_occupations = NULL,
      outcome = outcome
    )))
  }
}
```

```
# Second stage: each occupation's own Mother:Post estimate regressed on its exposure
  ↳ ,
# precision-weighted.
occ_stats <- bind_rows(lapply(exposure_index$occupation_code, function(code) {
  df_occ <- filter(df_ddd, MishlachYad_ISCO_08_2 == code)
  fit <- tryCatch({
    out <- capture.output(res <- suppressWarnings(run_intensive_margin_reg(df_occ))
    ↳ )
    res
  }, error = function(e) NULL)
  if (is.null(fit) || !"Mother:Post" %in% names(coef(fit$models$hours))) {
    return(tibble(occupation_code = code, beta_j = NA_real_, se_j = NA_real_))
  }
  m <- fit$models$hours
  tibble(
    occupation_code = code,
    beta_j = unname(coef(m)[["Mother:Post"]]),
    se_j = unname(se(m)[["Mother:Post"]])
  )
}))

joined <- exposure_index %>% left_join(occ_stats, by = "occupation_code")
is_dropped <- is.na(joined$beta_j) | is.na(joined$se_j) | joined$se_j <= 0
mechanism_df <- joined[!is_dropped, ]
dropped_df <- joined[is_dropped, ]

n_dropped <- nrow(dropped_df)
if (n_dropped > 0) {
  message(n_dropped, " of ", nrow(exposure_index), " occupation(s) dropped from the
  ↳ hours ",
    "mechanism regression (Mother:Post or its SE could not be estimated --
  ↳ insufficient ",
    "data or a degenerate fit).")
  message(sprintf(
    " wfh_exposure -- dropped occupations: mean = %.3f (n = %d); retained
    ↳ occupations: mean = %.3f (n = %d).",
    mean(dropped_df$wfh_exposure), n_dropped,
    mean(mechanism_df$wfh_exposure), nrow(mechanism_df)
  ))
})

reg_mechanism <- lm(beta_j ~ wfh_exposure, data = mechanism_df, weights = 1 / se_j
  ↳ ^2)
print(summary(reg_mechanism))

invisible(list(
  table = table_ddd,
  model = reg_ddd,
  n_employed = n_employed,
  n_matched = n_matched,
  n_clusters = n_clusters,
  # The regressor's values on the estimation sample, for the per-SD MDE.
  exposure_vector = df_ddd$WFH_Exposure,
  models = list(ddd = reg_ddd, mechanism = reg_mechanism),
  mechanism_data = mechanism_df,
  dropped_occupations = list(
    data = dropped_df,
    n_dropped = n_dropped,
    mean_exposure_dropped = if (n_dropped > 0) mean(dropped_df$wfh_exposure) else
    ↳ NA_real_,
    mean_exposure_retained = mean(mechanism_df$wfh_exposure)
  ),
  outcome = outcome
))
}
```

scripts/hours_ddd_saturated.R

```
# hours_ddd_saturated.R
# Saturated hours DDD: occupation x year, occupation x mother and mother x year fixed
  ↳ effects
# absorb every lower-order term of the pooled specification, so only the triple
  ↳ interaction
# carries a functional form. Mother:Post is absorbed by Mother`ShnatSeker and has no
  ↳ counterpart
# here. Same join, sample and clustering as run_hours_ddd_regression().
library(tidyverse)
```

```
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))

run_hours_ddd_saturated <- function(cleaned_df, exposure_index, controls = DEFAULT_
  ↪ CONTROLS,
                                outcome = "WorkHoursCont") {

  n_employed <- sum(cleaned_df$Employed == 1, na.rm = TRUE)

  df_ddd <- cleaned_df %>%
    filter(Employed == 1) %>%
    inner_join(
      exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
        ↪ = wfh_exposure),
      by = "MishlachYad_ISCO_08_2"
    )

  n_matched <- nrow(df_ddd)
  message(sprintf(
    "run_hours_ddd_saturated: %d of %d employed rows (%.1f%%) retained an occupation-
      ↪ level WFH_Exposure match.",
    n_matched, n_employed, 100 * n_matched / n_employed
  ))

  fixed_effects <- c("MishlachYad_ISCO_08_2`ShnatSeker", "MishlachYad_ISCO_08_2`
    ↪ Mother",
                    "Mother`ShnatSeker")

  formula_sat <- as.formula(paste(
    outcome, "~ Mother:Post:WFH_Exposure +", paste(controls, collapse = " + "),
    "|", paste(fixed_effects, collapse = " + ")
  ))

  reg_sat <- feols(formula_sat, data = df_ddd, cluster = ~MishlachYad_ISCO_08_2)
  check_for_dropped_coefficients(reg_sat, "run_hours_ddd_saturated()’s triple
    ↪ interaction")

  if (!"Mother:Post:WFH_Exposure" %in% names(coef(reg_sat))) {
    stop("run_hours_ddd_saturated: the triple interaction was not estimated -- it is
      ↪ collinear ",
         "with the fixed effects on this sample.")
  }

  table_sat <- etable(reg_sat, headers = c(paste0(outcome, " (hours DDD, saturated FE
    ↪ )),),
                    digits = 4)

  print(table_sat)

  invisible(list(
    table = table_sat,
    model = reg_sat,
    n_employed = n_employed,
    n_matched = n_matched,
    n_clusters = n_distinct(df_ddd$MishlachYad_ISCO_08_2),
    fixed_effects = fixed_effects
  ))
}
```

scripts/hours_ddd_binned.R

```
# hours_ddd_binned.R
# Quartile-binned hours DDD: the exposure score replaced by the pre-period quartile it
  ↪ falls in
# (the dose-response figure’s bins, passed in as breaks), so the top-us-bottom
  ↪ contrast carries a
# standard error. Coefficients are extracted by name. Bins are worker-weighted and
  ↪ hold unequal
# numbers of occupations, which is the cluster, so the per-quartile occupation counts
  ↪ are returned.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))
source(file.path("scripts", "assign_wfh_quartile.R"))

run_hours_ddd_binned <- function(cleaned_df, exposure_index, breaks, controls =
```

```
  ↪ DEFAULT_CONTROLS,
                                ref_quartile = 1, outcome = "WorkHoursCont") {

  if (length(breaks) != 5 || any(duplicated(breaks))) {
    stop("run_hours_ddd_binned: ‘breaks’ must be five strictly increasing quartile
      ↪ edges ",
         "(build_hours_dose_response()$breaks or compute_occupation_exposure_breaks()
      ↪ ).")
  }

  n_employed <- sum(cleaned_df$Employed == 1, na.rm = TRUE)

  df_ddd <- cleaned_df %>%
    filter(Employed == 1) %>%
    inner_join(
      exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
        ↪ = wfh_exposure),
      by = "MishlachYad_ISCO_08_2"
    ) %>%
    assign_wfh_quartile(breaks)

  n_matched <- nrow(df_ddd)
  message(sprintf(
    "run_hours_ddd_binned: %d of %d employed rows (%.1f%%) retained an occupation-
      ↪ level WFH_Exposure match.",
    n_matched, n_employed, 100 * n_matched / n_employed
  ))

  # Occupations and estimation rows per quartile.
  quartile_sizes <- df_ddd %>%
    filter(!is.na(data[[outcome]])) %>%
    group_by(WFH_Exposure_Q) %>%
    summarise(
      n_occupations = n_distinct(MishlachYad_ISCO_08_2),
      n_rows = n(),
      exposure_low = min(WFH_Exposure),
      exposure_high = max(WFH_Exposure),
      groups = "drop"
    ) %>%
    arrange(WFH_Exposure_Q)
  message("run_hours_ddd_binned: occupations and rows per exposure quartile:")
  print(as.data.frame(quartile_sizes))

  formula_bin <- as.formula(paste(
    outcome, sprintf("~ Mother * Post * i(WFH_Exposure_Q, ref = %d) +", ref_quartile)
    ↪ ,
    paste(controls, collapse = " + ")
  ))

  reg_bin <- feols(formula_bin, data = df_ddd, cluster = ~MishlachYad_ISCO_08_2)
  check_for_dropped_coefficients(reg_bin, "run_hours_ddd_binned()’s quartile triple
    ↪ interactions")

  triple_pattern <- ~"Mother:Post:WFH_Exposure_Q::(\\d)$"
  triple_terms <- grep(triple_pattern, names(coef(reg_bin)), value = TRUE)
  if (length(triple_terms) == 0) {
    stop("run_hours_ddd_binned: no Mother:Post:WFH_Exposure_Q::<q> coefficients found
      ↪ . Names present: ",
         paste(names(coef(reg_bin)), collapse = " , "))
  }

  df_t <- degrees_freedom(reg_bin, type = "t")
  coefs <- tibble(
    term = triple_terms,
    quartile = as.integer(sub(triple_pattern, "\\1", triple_terms)),
    estimate = unname(coef(reg_bin)[triple_terms]),
    std_error = unname(se(reg_bin)[triple_terms])
  ) %>%
  mutate(
    t_stat = estimate / std_error,
    p_value = 2 * pt(abs(t_stat), df = df_t, lower.tail = FALSE),
    ci_low = estimate - qt(0.975, df = df_t) * std_error,
    ci_high = estimate + qt(0.975, df = df_t) * std_error,
    ref_quartile = ref_quartile
  ) %>%
  left_join(quartile_sizes, by = c("quartile" = "WFH_Exposure_Q")) %>%
  arrange(quartile)

  table_bin <- etable(reg_bin, headers = c(paste0(outcome, " (hours DDD, exposure
    ↪ quartiles)")),
```

```
                                digits = 4)

  print(table_bin)

  message("=== Hours DDD, quartile bins: Mother x Post x (quartile q vs quartile ",
    ↪ ref_quartile, ") ===")
  print(as.data.frame(coefs %>% select(quartile, estimate, std_error, p_value, ci_low
    ↪ , ci_high,
                                n_occupations, n_rows)), digits = 4)

  invisible(list(
    table = table_bin,
    model = reg_bin,
    coefs = coefs,
    quartile_sizes = quartile_sizes,
    breaks = breaks,
    n_employed = n_employed,
    n_matched = n_matched,
    n_clusters = n_distinct(df_ddd$MishlachYad_ISCO_08_2)
  ))
}
```

Event studies

scripts/hours_diagnostics.R

```
# hours_diagnostics.R
# Hours counterpart of run_diagnostics(): 2x2 cell means and the Mother x year event
  ↪ study on the
# employed, whose pre-period coefficients run_pretrend_joint_test() tests jointly.
library(tidyverse)
library(fixest)
source(file.path("scripts", "tidy_event_study_coefs.R"))

run_hours_diagnostics <- function(cleaned_df) {
  hours_df <- filter(cleaned_df, Employed == 1)

  # 1. Mean hours by Mother/Post cell
  message("=== 2x2 DiD mean weekly hours (Employed == 1) ===")
  hours_by_period <- hours_df %>%
    group_by(Mother, Post) %>%
    # n counts the rows the mean uses.
    summarise(mean_hours = mean(WorkHoursCont, na.rm = TRUE),
              n = sum(!is.na(WorkHoursCont)), groups = "drop")
  print(hours_by_period)

  # 2. Event study: i(ShnatSeker, ref = 2019) supplies the year effects; 2019 is the
    ↪ reference in
# both i() terms.
  message("=== Pre-trend test (event study, hours) ===")

  reg_pretrend_hours <- feols(
    as.formula(paste(
      "WorkHoursCont ~ Mother + i(ShnatSeker, ref = 2019) + i(ShnatSeker, Mother, ref
        ↪ = 2019) +",
      paste(DEFAULT_CONTROLS, collapse = " + ")
    )),
    data = hours_df, cluster = ~IDPUF
  )

  pretrend_table_hours <- etable(reg_pretrend_hours, digits = 4)
  print(pretrend_table_hours)

  # Tidy one-row-per-year frame of the Mother x year terms, selected by anchored name.
  pretrend_coefs_hours <- tidy_event_study_coefs(
    reg_pretrend_hours, term_suffix = "Mother", ref_year = 2019
  )

  print(as.data.frame(pretrend_coefs_hours %>%
    select(year, estimate, std_error, p_value, ci_low, ci_high,
      ↪ period)))

  invisible(list(
    hours_by_period = hours_by_period,
    pretrend_table = pretrend_table_hours,
    pretrend_coefs = pretrend_coefs_hours,
    pretrend_model = reg_pretrend_hours
  ))
}
```



```

}

scripts/hours_ddd_event_study.R

# hours_ddd_event_study.R
# Event-study form of the hours DDD: Post replaced by survey-year interactions at
#   ↳ every level
# (year, year x Mother, year x exposure, year x Mother x exposure), so the triple
#   ↳ difference is
# recovered year by year against 2019. The DiD-level pre-trend tests average over
#   ↳ exposure; this
# tests the pre-period coefficients of the triple interaction itself. MotherWPH is a
#   ↳ materialized
# product because i(f, var) takes a single variable.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))
source(file.path("scripts", "tidy_event_study_coefs.R"))

# term_suffix and ref_year are returned so the Wald test's keep regex is built from
#   ↳ them.
run_hours_ddd_event_study <- function(cleaned_df, exposure_index, controls = DEFAULT_CONTROLS,
                                     ref_year = 2019) {

  n_employed <- sum(cleaned_df$Employed == 1, na.rm = TRUE)

  # Same join and sample as run_hours_ddd_regression().
  df_es <- cleaned_df %>%
    filter(Employed == 1) %>%
    inner_join(
      exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WPH_Exposure
        ↳ = wph_exposure),
      by = "MishlachYad_ISCO_08_2"
    ) %>%
    mutate(MotherWPH = Mother * WPH_Exposure)

  n_matched <- nrow(df_es)
  message(sprintf(
    "run_hours_ddd_event_study: %d of %d employed rows (%.1f%%) retained an
    ↳ occupation-level WPH_Exposure match (dropped: disclosure-masked or
    ↳ unmapped ISCO codes).",
    n_matched, n_employed, 100 * n_matched / n_employed
  ))

  formula_es <- as.formula(paste(
    "WorkHoursCont ~ Mother * WPH_Exposure",
    sprintf("i(ShnatSeker, ref = %d)", ref_year),
    sprintf("i(ShnatSeker, Mother, ref = %d)", ref_year),
    sprintf("i(ShnatSeker, WPH_Exposure, ref = %d)", ref_year),
    sprintf("i(ShnatSeker, MotherWPH, ref = %d)", ref_year),
    paste(controls, collapse = " + "),
    sep = " + "
  ))

  # Clustered on occupation, as the pooled DDD is; about 40 clusters.
  reg_es <- feols(formula_es, data = df_es, cluster = ~MishlachYad_ISCO_08_2)
  check_for_dropped_coefficients(reg_es, "run_hours_ddd_event_study()'s triple-
    ↳ interaction event study")

  table_es <- etable(reg_es, headers = c("WorkHoursCont (hours DDD event study)",
    ↳ digits = 4)
  print(table_es)

  # Tidy frame of the triple-interaction terms, one row per non-reference year.
  term_suffix <- "MotherWPH"
  coefs <- tidy_event_study_coefs(reg_es, term_suffix = term_suffix, ref_year = ref_
    ↳ year)

  message("==== Hours DDD event study: Mother x year x WPH_Exposure (ref = ", ref_year
    ↳ ", ") =====")
  print(as.data.frame(coefs %>% select(year, estimate, std_error, p_value, ci_low, ci
    ↳ _high, period)))

  invisible(list(
    table = table_es,

```

```

    model = reg_es,
    coefs = coefs,
    ref_year = ref_year,
    term_suffix = term_suffix,
    n_employed = n_employed,
    n_matched = n_matched
  ))
}

Selection bounds

scripts/lee_trim_proportion.R

# lee_trim_proportion.R
# Lee (2009) trim proportion for the (Mother = 1, Post = 1) cell: the excess of the
#   ↳ observed
# selection rate s11 over the parallel-trends counterfactual s10 + (s01 - s00).

lee_trim_proportion <- function(s00, s01, s10, s11) {
  s11_counterfactual <- s10 + (s01 - s00)
  excess <- s11 > s11_counterfactual
  trim_prop <- if (excess) 1 - s11_counterfactual / s11 else 0
  list(s11_counterfactual = s11_counterfactual, excess_selection = excess, trim_prop
    ↳ = trim_prop)
}


```

scripts/lee_trim_cell.R

```

# lee_trim_cell.R
# Trims floor(trim_prop * n) rows from the top (lower bound) or bottom (upper bound)
#   ↳ of one cell's
# outcome distribution. Ties are broken by row order; callers must drop missing
#   ↳ outcomes first.
library(tidyverse)

lee_trim_cell <- function(cell_df, trim_prop, outcome_col = "WorkHoursCont") {
  n_cell <- nrow(cell_df)
  if (n_cell == 0 || trim_prop <= 0) {
    return(list(lower = cell_df, upper = cell_df, n_cell = n_cell, n_trim = 0L))
  }
  n_trim <- as.integer(floor(trim_prop * n_cell))
  ranked <- arrange(cell_df, data[[outcome_col]])
  # Indices are injected with !! so n_cell cannot resolve to a column of the same name
  #   ↳
  keep_lower <- seq_len(max(n_cell - n_trim, 0))
  keep_upper <- if (n_trim < n_cell) (n_trim + 1):n_cell else integer(0)
  list(lower = slice(ranked, !!keep_lower),
    upper = slice(ranked, !!keep_upper),
    n_cell = n_cell, n_trim = n_trim)
}


```

scripts/imbens_manski_ci.R

```

# imbens_manski_ci.R
# Imbens and Manski (2004) confidence interval for a partially identified parameter:
#   ↳ solve
#  $\Phi(c + \delta / \max(se)) - \Phi(-c) = \text{conf\_level}$  and report  $[\theta_{\text{eta}_L} - c \text{ se}_L, \theta_{\text{eta}_U} + c \text{ se}_U]$ ;
#   ↳
# collapses to the two-sided interval when  $\delta = 0$ .
imbens_manski_ci <- function(theta_L, theta_U, se_L, se_U, conf_level = 0.95) {
  delta <- max(theta_U - theta_L, 0)
  denom <- max(se_L, se_U)
  if (delta == 0 || !is.finite(denom) || denom <= 0) {
    z_ci <- qnorm(1 - (1 - conf_level) / 2)
    return(list(c_alpha = z_ci, lower = theta_L - z_ci * se_L, upper = theta_U + z_ci
      ↳ * se_U))
  }
  target <- function(c) pnorm(c + delta / denom) - pnorm(-c) - conf_level
  c_alpha <- uniroot(target, interval = c(0, 20), tol = 1e-10)$root
}


```

```

list(c_alpha = c_alpha, lower = theta_L - c_alpha * se_L, upper = theta_U + c_alpha
  ↳ * se_U)
}


```

scripts/intensive_margin_lee_bounds.R

```

# intensive_margin_lee_bounds.R
# Lee (2009) trimming bounds for the hours DiD. Employment is itself an outcome, so
#   ↳ the employed
# post-period mothers may differ in composition from the pre-period ones. Selection
#   ↳ rates
# s_ab = P(hours observed | Mother = a, Post = b) give the parallel-trends
#   ↳ counterfactual
# s11* = s10 + (s01 - s00); if s11 exceeds it, the excess share is trimmed from the
#   ↳ top (lower
# bound) or bottom (upper bound) of the (1, 1) cell's hours distribution and Mother:
#   ↳ Post is
# re-estimated. Handles excess selection in that cell only, under monotone selection;
#   ↳ rates are
# unweighted like every regression here.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "imbens_manski_ci.R"))
source(file.path("scripts", "lee_trim_proportion.R"))
source(file.path("scripts", "lee_trim_cell.R"))

run_intensive_margin_lee_bounds <- function(cleaned_df, controls = DEFAULT_CONTROLS)
  ↳ {

  # The rate is the share whose hours are observed (reference-week workers), not the
  #   ↳ share employed.
  sel_rates <- cleaned_df %>%
    group_by(Mother, Post) %>%
    summarise(selection_rate = mean(!is.na(WorkHoursCont)), n = n(), .groups = "drop"
      ↳ )

  get_rate <- function(m, p) {
    rate <- sel_rates$selection_rate[sel_rates$Mother == m & sel_rates$Post == p]
    if (length(rate) == 0) {
      stop("run_intensive_margin_lee_bounds: no rows found for Mother == ", m, ", ",
        ↳ Post == ", p,
        ↳ " -- all four (Mother, Post) cells must be present to compute the
        ↳ selection ",
        ↳ "counterfactual.")
    }
    rate
  }

  s00 <- get_rate(0, 0); s01 <- get_rate(0, 1); s10 <- get_rate(1, 0); s11 <- get_
    ↳ rate(1, 1)
  tp <- lee_trim_proportion(s00, s01, s10, s11)
  s11_counterfactual <- tp$s11_counterfactual
  excess <- tp$excess_selection
  trim_prop <- tp$trim_prop

  message(sprintf(
    paste(
      "run_intensive_margin_lee_bounds: selection (observed-hours) rates -- ",
      "Mother=0,Post=0: %.4f | Mother=0,Post=1: %.4f | Mother=1,Post=0: %.4f | ",
      "Mother=1,Post=1: %.4f (parallel-trends counterfactual: %.4f).\n %s"
    ),
    s00, s01, s10, s11, s11_counterfactual,
    if (excess) {
      sprintf("Excess selection detected in Mother=1,Post=1: trimming %.2f%% of that
        ↳ cell.",
        100 * trim_prop)
    } else {
      paste("No excess selection in Mother=1,Post=1 relative to the parallel-trends",
        ↳ "counterfactual -- bounds collapse to the untrimmed point estimate.")
    }
  ))

  rhs
    ↳ <- paste("Mother + Post + Mother:Post", paste(controls, collapse = "
    ↳ ", sep = " + "))
  formula_hours <- as.formula(paste("WorkHoursCont ~", rhs))
  fit_on <- function(df) feols(formula_hours, data = df, cluster = ~IDPUF)
}


```

```
# Restrict to rows with an observed outcome before trimming: arrange() sorts NA last
↳ .
employed_df <- filter(cleaned_df, Employed == 1, !is.na(WorkHoursCont))
point_reg <- fit_on(employed_df)

if (!excess || trim_prop == 0) {
  lower_reg <- point_reg
  upper_reg <- point_reg
  n_trimmed <- 0L
} else {
  treated_cell <- filter(employed_df, Mother == 1, Post == 1)
  other_cells <- filter(employed_df, !(Mother == 1 & Post == 1))
  # Lower bound drops the highest-hours rows, upper bound the lowest (lee_trim_cell.
  ↳ R).
  trimmed <- lee_trim_cell(treated_cell, trim_prop)

  lower_reg <- fit_on(bind_rows(other_cells, trimmed$lower))
  upper_reg <- fit_on(bind_rows(other_cells, trimmed$upper))
  n_trimmed <- trimmed$n_trim
}

# Mother:Post is dropped for collinearity if the whole (1, 1) cell is trimmed away;
↳ fail
# informatively.
extract <- function(m, fn) {
  val <- fn(m)[["Mother:Post"]]
  if (is.na(val)) {
    stop("run_intensive_margin_lee_bounds: Mother:Post is not estimable in this
    ↳ bound's trimmed ",
    "sample (dropped for collinearity) -- this happens when the entire Mother
    ↳ =1,Post=1 cell ",
    "has been trimmed away (trim_prop == 1), leaving no data to identify the
    ↳ interaction.")
  }
  unname(val)
}
co <- function(m) extract(m, coef)
se_of <- function(m) extract(m, se)

se_lower <- se_of(lower_reg)
se_point <- se_of(point_reg)
se_upper <- se_of(upper_reg)

z <- qnorm(0.975)
bounds_table <- tibble(
  bound = c("lower", "point (untrimmed)", "upper"),
  mother_post_coef = c(co(lower_reg), co(point_reg), co(upper_reg)),
  se = c(se_lower, se_point, se_upper),
  ci_low = c(co(lower_reg) - z * se_lower, co(point_reg) - z * se_point,
    ↳ co(upper_reg) - z * se_upper),
  ci_high = c(co(lower_reg) + z * se_lower, co(point_reg) + z * se_point,
    ↳ co(upper_reg) + z * se_upper)
)
print(as.data.frame(bounds_table), digits = 4)

# Imbens-Manski interval for the identified set.
im_ci <- imbens_manski_ci(co(lower_reg), co(upper_reg), se_lower, se_upper)
message(sprintf(
  "run_intensive_margin_lee_bounds: 95%% Imbens-Manski CI for the identified set =
  ↳ [%f, %f] (c.alpha = %f).",
  im_ci$lower, im_ci$upper, im_ci$c_alpha
))

return(invisible(list(
  table = bounds_table,
  models = list(point = point_reg, lower = lower_reg, upper = upper_reg),
  diagnostics = list(
    selection_rates = sel_rates,
    s11_counterfactual = s11_counterfactual,
    excess_selection = excess,
    trim_prop = trim_prop,
    n_trimmed = n_trimmed
  ),
  imbens_manski_ci = im_ci
)))
})
```

scripts/hours_ddd_lee_bounds.R

```
# hours_ddd_lee_bounds.R
# Lee bounds for the hours DDD, generalized from intensive_margin_lee_bounds.R. The
↳ regressor is
# the occupation-level score, undefined for the non-employed, so the selection
↳ counterfactual is
# computed within quartiles of the pre-period demographic-cell index, which is defined
↳ for
# everyone. Each quartile is trimmed on its own excess share and the triple-
↳ interaction formula is
# refit on the recombined samples. The point fit here additionally requires a matched
↳ cell, so it
# can differ slightly from run_hours_ddd_regression()'s.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "imbens_manski_ci.R"))
source(file.path("scripts", "lee_trim_proportion.R"))
source(file.path("scripts", "lee_trim_cell.R"))
source(file.path("robustness", "age_balance_robustness.R")) # compute_pre_period_
↳ quartile_breaks()

MIN_CELL_WARN <- 30

run_hours_ddd_lee_bounds <- function(cleaned_df, exposure_index, exposure_cells,
  controls = DEFAULT_CONTROLS) {

  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
  breaks <- compute_pre_period_quartile_breaks(cleaned_df, exposure_cells)

  # Full sample with a matched cell index, quartile-assigned: what the selection rates
  ↳ use.
  full_df <- cleaned_df %>%
    left_join(exposure_cells, by = exposure_join_vars) %>%
    filter(!is.na(WFH_Exposure)) %>%
    assign_wfh_quartile(breaks) %>%
    rename(WFH_Exposure_Cell = WFH_Exposure)

  quartiles <- sort(unique(full_df$WFH_Exposure_Q))

  # Share whose hours are observed, not share employed.
  get_rate <- function(m, p, q) {
    sub <- full_df$WorkHoursCont[full_df$Mother == m & full_df$Post == p &
      full_df$WFH_Exposure_Q == q]
    if (length(sub) == 0) {
      stop("run_hours_ddd_lee_bounds: no rows found for Mother == ", m, ", Post == ",
      ↳ p,
      " WFH_Exposure_Q == ", q, " -- all four (Mother, Post) cells must be
      ↳ present in ",
      "every quartile to compute the selection counterfactual.")
    }
    mean(!is.na(sub))
  }

  quartile_diag <- map_dfr(quartiles, function(q) {
    s00 <- get_rate(0, 0, q); s01 <- get_rate(0, 1, q); s10 <- get_rate(1, 0, q); s11
    ↳ <- get_rate(1, 1, q)
    n11 <- sum(full_df$Mother == 1 & full_df$Post == 1 & full_df$WFH_Exposure_Q == q)
    tp <- lee_trim_proportion(s00, s01, s10, s11)
    tibble(WFH_Exposure_Q = q, s00 = s00, s01 = s01, s10 = s10, s11 = s11,
      s11_counterfactual = tp$s11_counterfactual, n_mother1_post1 = n11,
      excess_selection = tp$excess_selection, trim_prop = tp$trim_prop)
  })

  message("run_hours_ddd_lee_bounds: selection (observed-hours) rates and trim
  ↳ proportions by WFH_Exposure quartile:")
  print(as.data.frame(quartile_diag), digits = 4)

  thin <- filter(quartile_diag, n_mother1_post1 < MIN_CELL_WARN)
  if (nrow(thin) > 0) {
    warning(sprintf(
      "run_hours_ddd_lee_bounds: %d quartile(s) have fewer than %d Mother=1,Post=1
      ↳ rows (%s) -- their trim_prop/bounds may be noisy.",
      nrow(thin), MIN_CELL_WARN, paste(sprintf("Q%d: n=%d", thin$WFH_Exposure_Q, thin
      ↳ $n_mother1_post1), collapse = "; ")
    ))
  }

  # Employed, occupation-matched, observed-hours sample carrying each row's cell
```

```
↳ quartile.
employed_df <- full_df %>%
  filter(Employed == 1, !is.na(WorkHoursCont)) %>%
  inner_join(
    exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
    ↳ = wfh_exposure),
    by = "MishlachYad_ISCO_08_2"
  )

other_cells <- filter(employed_df, !(Mother == 1 & Post == 1))

# Each quartile is trimmed independently on its own trim_prop (lee_trim_cell.R).
trim_by_quartile <- map(quartiles, function(q) {
  lee_trim_cell(filter(employed_df, Mother == 1, Post == 1, WFH_Exposure_Q == q),
    trim_prop = quartile_diag$trim_prop[quartile_diag$WFH_Exposure_Q ==
    ↳ q])
})

n_trimmed_by_quartile <- tibble(
  WFH_Exposure_Q = quartiles,
  n_cell = map_int(trim_by_quartile, "n_cell"),
  n_trimmed = map_int(trim_by_quartile, "n_trim")
)

lower_df <- bind_rows(other_cells, bind_rows(map(trim_by_quartile, "lower")))
upper_df <- bind_rows(other_cells, bind_rows(map(trim_by_quartile, "upper")))

rhs <- paste("Mother*Post*WFH_Exposure", paste(controls, collapse = " + "), sep
↳ = " + ")
formula_hours <- as.formula(paste("WorkHoursCont ~", rhs))
# Clustered on occupation, as the headline is.
fit_on <- function(df) feols(formula_hours, data = df, cluster = ~MishlachYad_ISCO_
↳ 08_2)

point_reg <- fit_on(employed_df)
lower_reg <- fit_on(lower_df)
upper_reg <- fit_on(upper_df)

co <- function(m) unname(coef(m)[["Mother:Post:WFH_Exposure"]])
se_of <- function(m) unname(se(m)[["Mother:Post:WFH_Exposure"]])

se_lower <- se_of(lower_reg); se_point <- se_of(point_reg); se_upper <- se_of(upper
↳ reg)
z <- qnorm(0.975)
bounds_table <- tibble(
  bound = c("lower", "point (untrimmed)", "upper"),
  coef = c(co(lower_reg), co(point_reg), co(upper_reg)),
  se = c(se_lower, se_point, se_upper),
  ci_low = c(co(lower_reg) - z * se_lower, co(point_reg) - z * se_point, co(upper
    ↳ reg) - z * se_upper),
  ci_high = c(co(lower_reg) + z * se_lower, co(point_reg) + z * se_point, co(upper
    ↳ reg) + z * se_upper),
  n_obs = c(nobs(lower_reg), nobs(point_reg), nobs(upper_reg))
)
print(as.data.frame(bounds_table), digits = 4)

im_ci <- imbens_manski_ci(co(lower_reg), co(upper_reg), se_lower, se_upper)
message(sprintf(
  "run_hours_ddd_lee_bounds: 95%% Imbens-Manski CI for Mother:Post:WFH_Exposure's
  ↳ identified set = [%f, %f] (c.alpha = %f).",
  im_ci$lower, im_ci$upper, im_ci$c_alpha
))

invisible(list(
  table = bounds_table,
  models = list(point = point_reg, lower = lower_reg, upper = upper_reg),
  diagnostics = list(
    quartile_selection_rates = quartile_diag,
    n_trimmed_by_quartile = n_trimmed_by_quartile,
    n_trimmed_total = sum(n_trimmed_by_quartile$n_trimmed)
  ),
  imbens_manski_ci = im_ci
))
}
```

Inference

robustness/hours_ddd_inference.R

```
# hours_ddd_inference.R
# Small-cluster inference for the hours DDD, whose regressor varies across forty
#   ↳ occupations.
# run_hours_ddd_wild_bootstrap(): wild cluster bootstrap p-values (Rademacher, null
#   ↳ imposed) via
# fwildclusterboot::boottest(), for the headline and the event study's pre-period
#   ↳ terms; the joint
# pre-trend test stays the analytic Wald F. run_hours_ddd_permutation_test(): the
#   ↳ forty exposure
# scores are reassigned across occupations, the DDD refit and the cluster-robust t
#   ↳ recorded; the
# studentized statistic is used because the raw coefficient is not pivotal across
#   ↳ reassignments.
# fwildclusterboot is loaded inside the function; dqrng::dqset.seed() is what fixes
#   ↳ its draws.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))

run_hours_ddd_wild_bootstrap <- function(model, param, clustid = "MishlachYad_ISCO_08
  ↳ _2",
                                     B = 9999, seed = 20260922, R = NULL, r = 0,
                                     type = "rademacher", label = NULL) {
  for (pkg in c("fwildclusterboot", "dqrng")) {
    if (!requireNamespace(pkg, quietly = TRUE)) {
      stop("run_hours_ddd_wild_bootstrap: package '", pkg, "' is not installed. It is
        ↳ the one ",
        "dependency beyond tidyverse + fixest (see README.md); install it or skip
        ↳ this step.")
    }
  }
  if (!all(param %in% names(coef(model)))) {
    stop("run_hours_ddd_wild_bootstrap: coefficient(s) not in the model: ",
         paste(setdiff(param, names(coef(model))), collapse = ", "))
  }

  dqrng::dqset.seed(seed)
  set.seed(seed)

  args <- list(object = model, param = param, clustid = clustid, B = B, type = type,
               impose_null = TRUE)
  if (!is.null(R)) {
    args$R <- R
    args$r <- r
  }

  # The package's one-time reproducibility note is suppressed.
  bt <- suppressWarnings(suppressMessages(do.call(fwildclusterboot::boottest, args)))

  ci <- bt$conf_int
  if (is.null(ci) || length(ci) < 2) ci <- c(NA_real_, NA_real_)

  tbl <- tibble(
    label      = if (is.null(label)) paste(param, collapse = " + ") else label,
    param      = paste(param, collapse = " + "),
    estimate    = unname(bt$point_estimate),
    t_stat      = unname(bt$t_stat),
    p_boot      = unname(bt$p_val),
    ci_low      = unname(ci[1]),
    ci_high     = unname(ci[2]),
    B          = as.integer(B),
    n_clusters  = as.integer(bt$n_G),
    weights     = type,
    null_imposed = TRUE,
    seed       = as.integer(seed)
  )

  message(sprintf(
    "run_hours_ddd_wild_bootstrap [%s]: estimate = %.4f, t = %.3f, bootstrap p = %.4f
      ↳ (B = %d, %d clusters, %s weights).",
    tbl$label, tbl$estimate, tbl$t_stat, tbl$p_boot, B, tbl$n_clusters, type
  ))

  invisible(list(table = tbl, boottest = bt))
}

run_hours_ddd_permutation_test <- function(cleaned_df, exposure_index, n_perm = 999,
  ↳ controls,
  coef_name = "Mother:Post:WFH_Exposure",
  outcome = "WorkHoursCont", report_every =
    ↳ 100) {
  if (n_perm < 1) stop("run_hours_ddd_permutation_test: n_perm must be at least 1.")

  # Same join as run_hours_ddd_regression(), so the observed statistic is the headline
  ↳ 's.
  df_ddd <- cleaned_df %>%
    filter(Employed == 1) %>%
    inner_join(
      exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
        ↳ = wfh_exposure),
      by = "MishlachYad_ISCO_08_2"
    )

  formula_ddd <- as.formula(paste(
    outcome, "~ Mother*Post*WFH_Exposure +", paste(controls, collapse = " + ")
  ))
  fit <- function(d) feols(formula_ddd, data = d, cluster = ~MishlachYad_ISCO_08_2,
    ↳ notes = FALSE)

  m_obs <- fit(df_ddd)
  if (!coef_name %in% names(coef(m_obs))) {
    stop("run_hours_ddd_permutation_test: '", coef_name, "' not estimated on the
      ↳ observed data.")
  }
  b_obs <- unname(coef(m_obs)[[coef_name]])
  t_obs <- b_obs / unname(se(m_obs)[[coef_name]])
  n_clusters <- n_distinct(df_ddd$MishlachYad_ISCO_08_2)

  # Permute the score vector and index it onto rows.
  codes <- exposure_index$occupation_code
  vals <- exposure_index$wfh_exposure
  idx <- match(df_ddd$MishlachYad_ISCO_08_2, codes)

  set.seed(seed)
  coef_perm <- numeric(n_perm)
  t_perm <- numeric(n_perm)
  started <- Sys.time()
  for (i in seq_len(n_perm)) {
    df_ddd$WFH_Exposure <- sample(vals)[idx]
    m <- fit(df_ddd)
    if (coef_name %in% names(coef(m))) {
      coef_perm[i] <- unname(coef(m)[[coef_name]])
      t_perm[i] <- coef_perm[i] / unname(se(m)[[coef_name]])
    } else {
      coef_perm[i] <- NA_real_
      t_perm[i] <- NA_real_
    }
    if (report_every > 0 && i %% report_every == 0) {
      elapsed <- as.numeric(difftime(Sys.time(), started, units = "secs"))
      message(sprintf("run_hours_ddd_permutation_test: %d / %d draws (%.0f s elapsed)
        ↳ ",
                      i, n_perm, elapsed))
    }
  }

  ok <- is.finite(t_perm)
  n_ok <- sum(ok)
  # Phipson and Smyth (2010): count the observed statistic among the draws.
  p_perm_t <- (1 + sum(abs(t_perm[ok]) >= abs(t_obs))) / (1 + n_ok)
  p_perm_coef <- (1 + sum(abs(coef_perm[ok]) >= abs(b_obs))) / (1 + n_ok)

  tbl <- tibble(
    coef_name = coef_name,
    estimate   = b_obs,
    t_stat     = t_obs,
    p_perm     = p_perm_t,
    p_perm_coef = p_perm_coef,
    n_perm     = as.integer(n_perm),
    n_perm_valid = as.integer(n_ok),
    seed       = as.integer(seed),
    n_clusters = as.integer(n_clusters),
    t_perm_q025 = unname(quantile(t_perm[ok], 0.025)),
    t_perm_q975 = unname(quantile(t_perm[ok], 0.975)),
    t_perm_max_abs = max(abs(t_perm[ok])),
    coef_perm_q025 = unname(quantile(coef_perm[ok], 0.025)),
    coef_perm_q975 = unname(quantile(coef_perm[ok], 0.975)),
    # Monte-Carlo standard error of p at the reported value, so a reader can see the
    ↳ resolution.
  )
}
```

```
p_perm_mc_se = sqrt(p_perm_t * (1 - p_perm_t) / n_ok)
)

draws <- tibble(draw = seq_len(n_perm), coef = coef_perm, t_stat = t_perm)

message(sprintf(
  paste0("run_hours_ddd_permutation_test: observed %s = %.4f (t = %.3f);
    ↳ permutation p = %.4f ",
    "on the |t| scale (%.4f on the |coef| scale), %d valid draws of %d, seed %
    ↳ d. ",
    "Largest |t| under the null: %.3f."),
  coef_name, b_obs, t_obs, p_perm_t, p_perm_coef, n_ok, n_perm, seed, tbl$t_perm
  ↳ max_abs
))

invisible(list(table = tbl, draws = draws, model = m_obs))
}
```

Calibration and sorting checks

scripts/hours_ddd_swap_control.R

```
# hours_ddd_swap_control.R
# Fixed-sample test of the calibration: the external score as regressor on all forty
#   ↳ occupations.
# with the ten swapped occupations given their own Mother a Post structure. The
#   ↳ unswapped-30 row
# changes the sample as well as the measure; this keeps the sample and lets Mother:
#   ↳ Post:Swapped
# absorb the swapped group's own post-2021 change in the motherhood gap.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))

run_hours_ddd_swap_control <- function(cleaned_df, external_index, swapped_codes,
  controls = DEFAULT_CONTROLS, outcome = "
  ↳ WorkHoursCont") {
  if (!outcome %in% names(cleaned_df)) {
    stop("run_hours_ddd_swap_control: outcome column '", outcome, "' is not in
      ↳ cleaned_df.")
  }
  if (!all(c("occupation_code", "wfh_exposure") %in% names(external_index))) {
    stop("run_hours_ddd_swap_control: external_index needs 'occupation_code' and 'wfh
      ↳ _exposure' columns.")
  }
  swapped_codes <- as.numeric(swapped_codes)
  if (length(swapped_codes) == 0) {
    stop("run_hours_ddd_swap_control: swapped_codes is empty -- with no swapped
      ↳ occupations the ",
      "Swapped indicator is constant and the specification collapses to the
      ↳ external-index DDD.")
  }

  n_employed <- sum(cleaned_df$Employed == 1, na.rm = TRUE)

  df_ddd <- cleaned_df %>%
    filter(Employed == 1) %>%
    inner_join(
      external_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
        ↳ = wfh_exposure),
      by = "MishlachYad_ISCO_08_2"
    ) %>%
    mutate(Swapped = as.integer(MishlachYad_ISCO_08_2 %in% swapped_codes))

  n_matched <- nrow(df_ddd)
  n_clusters <- n_distinct(df_ddd$MishlachYad_ISCO_08_2)
  n_swapped <- n_distinct(df_ddd$MishlachYad_ISCO_08_2[df_ddd$Swapped == 1])
  share_swapped_rows <- mean(df_ddd$Swapped)
  message(sprintf(
    paste0("run_hours_ddd_swap_control: %d of %d employed rows matched (%d
      ↳ occupations); %d swapped ",
    "occupations carry %.if%% of the rows."),
    n_matched, n_employed, n_clusters, n_swapped, 100 * share_swapped_rows
  ))
}
```

```

if (n_swapped == 0 || n_swapped == n_clusters) {
  stop("run_hours_ddd_swap_control: the Swapped indicator is constant on the
    ↳ matched sample ",
      "(", n_swapped, " of ", n_clusters, " occupations swapped).")
}

formula_swap <- as.formula(paste(
  outcome, "~ Mother * Post * WFH_Exposure + Mother * Post * Swapped +",
  paste(controls, collapse = " + ")
))
model <- feols(formula_swap, data = df_ddd, cluster = ~MishlachYad_ISCO_08_2)
check_for_dropped_coefficients(model, "run_hours_ddd_swap_control()"s triple
  ↳ interactions")

terms <- c("Mother:Post:WFH_Exposure", "Mother:Post:Swapped")
missing_terms <- setdiff(terms, names(coef(model)))
if (length(missing_terms) > 0) {
  stop("run_hours_ddd_swap_control: not estimated: ", paste(missing_terms, collapse
    ↳ = ", "))
}
coefs <- tibble(
  term = terms,
  estimate = unname(coef(model)[terms]),
  std_error = unname(se(model)[terms]),
  p_value = unname(pvalue(model)[terms]),
  n_clusters = n_clusters,
  n_swapped_occupations = n_swapped
)

table <- etable(model, headers = c(paste0(outcome, " (external index + swapped-
  ↳ occupation terms)"),
  digits = 4)
print(table)
message("=== External-score gradient with the swapped occupations' own Mother x
  ↳ Post change absorbed ===")
print(as.data.frame(coefs), digits = 4)

invisible(list(
  table = table,
  model = model,
  coefs = coefs,
  n_employed = n_employed,
  n_matched = n_matched,
  n_clusters = n_clusters,
  n_swapped_occupations = n_swapped,
  share_swapped_rows = share_swapped_rows,
  outcome = outcome
))
}

```

scripts/exposure_sorting_check.R

```

# exposure_sorting_check.R
# Sorting check: did mothers move into teleworkable occupations after 2021? A DiD with
  ↳ the
# exposure score itself as the outcome, on the hours DDD's sample: (i) mean exposure,
  ↳ (ii) a
# top-quartile indicator, (iii) the event-study form of (i). Clustered by individual,
  ↳ since the
# outcome varies at the individual level.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "tidy_event_study_coefs.R"))
source(file.path("scripts", "assign_wfh_quartile.R"))

run_exposure_sorting_check <- function(cleaned_df, exposure_index, breaks,
  controls = DEFAULT_CONTROLS, ref_year = 2019,
  outcome_sample = "WorkHoursCont") {
  if (length(breaks) != 5 || any(duplicated(breaks))) {
    stop("run_exposure_sorting_check: 'breaks' must be five strictly increasing
      ↳ quartile edges ",
        "(build_hours_dose_response()$breaks).")
  }

  # The DDD's estimation sample; outcome_sample = NULL widens it to all matched

```

```

  ↳ employed women.
df <- cleaned_df %>%
  filter(Employed == 1) %>%
  inner_join(
    exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
      ↳ = wfh_exposure),
    by = "MishlachYad_ISCO_08_2"
  )
if (!is.null(outcome_sample)) {
  if (!outcome_sample %in% names(df)) {
    stop("run_exposure_sorting_check: '", outcome_sample, "' is not a column; pass
      ↳ outcome_sample = NULL ",
        "to use all matched employed rows.")
  }
  df <- filter(df, !is.na(.data[outcome_sample]))
}
df <- df %>%
  assign_wfh_quartile(breaks) %>%
  mutate(TopQuartile = as.integer(WFH_Exposure_Q == 4L))

message(sprintf("run_exposure_sorting_check: %d rows, %d occupations, %d in the top
  ↳ quartile's occupations.",
    nrow(df), n_distinct(df$MishlachYad_ISCO_08_2),
    n_distinct(df$MishlachYad_ISCO_08_2[df$TopQuartile == 1L])))

rhs_did <- paste("Mother * Post +", paste(controls, collapse = " + "))
m_exposure <- feols(as.formula(paste("WFH_Exposure ~", rhs_did)), data = df,
  ↳ cluster = ~IDPUF)
m_top <- feols(as.formula(paste("TopQuartile ~", rhs_did)), data = df,
  ↳ cluster = ~IDPUF)

# Pre-period levels, so the DiD can be read against where each group started.
pre <- df %>%
  filter(Post == 0) %>%
  group_by(Mother) %>%
  summarise(mean_exposure = mean(WFH_Exposure), share_top = mean(TopQuartile),
    sd_exposure = sd(WFH_Exposure), n = n(), .groups = "drop")
pre_of <- function(m, col) { v <- pre[[col]][pre$Mother == m]; if (length(v) == 0)
  ↳ NA_real_ else v }

extract <- function(m, outcome_name, pre_col) {
  tibble(
    outcome = outcome_name,
    estimate = unname(coef(m)[["Mother:Post"]]),
    std_error = unname(se(m)[["Mother:Post"]]),
    p_value = unname(pvalue(m)[["Mother:Post"]]),
    n = as.integer(nobs(m)),
    pre_mean_mothers = pre_of(1, pre_col),
    pre_mean_childless = pre_of(0, pre_col),
    pre_sd_exposure = sd(df$WFH_Exposure[df$Post == 0])
  )
}
table <- bind_rows(
  extract(m_exposure, "Occupation-level WFH exposure (mean)", "mean_exposure"),
  extract(m_top, "In top exposure quartile (share)", "share_top")
) %>%
  mutate(estimate_per_pre_sd = if_else(outcome == "Occupation-level WFH exposure (
    ↳ mean)",
    estimate / pre_sd_exposure, NA_real_))

# Event-study form; skipped without a survey-year column.
es_coefs <- NULL
m_es <- NULL
if ("ShnatSeker" %in% names(df)) {
  m_es <- feols(
    as.formula(paste(
      sprintf("WFH_Exposure ~ Mother + i(ShnatSeker, ref = %d) + i(ShnatSeker,
        ↳ Mother, ref = %d) +",
        ref_year, ref_year),
      paste(controls, collapse = " + ")
    )),
    data = df, cluster = ~IDPUF
  )
  es_coefs <- tidy_event_study_coefs(m_es, term_suffix = "Mother", ref_year = ref_
    ↳ year)
} else {
  message("run_exposure_sorting_check: no ShnatSeker column -- the event-study
    ↳ version is skipped.")
}

```

```

message("=== Sorting check: DiD on occupational exposure itself (Mother x Post,
  ↳ clustered by IDPUF) ===")
print(as.data.frame(table), digits = 4)
if (!is.null(es_coefs)) {
  message("=== Sorting check, event-study form (Mother x year on exposure, ref ",
    ↳ ref_year, ") ===")
  print(as.data.frame(es_coefs %>% select(year, estimate, std_error, p_value)),
    ↳ digits = 4)
}

invisible(list(
  table = table,
  pre_levels = pre,
  event_study = es_coefs,
  models = list(exposure = m_exposure, top_quartile = m_top, event_study = m_
    ↳ es),
  ref_year = ref_year
))
}

```

scripts/hours_ddd_cell_exposure.R

```

# hours_ddd_cell_exposure.R
# The hours DDD on the pre-period demographic-cell index, which a woman's later
  ↳ occupation cannot
# move: a bound on occupational sorting. The cell index is coarser and far less
  ↳ dispersed than the
# occupation score, so only the per-SD figure is comparable. Clustered on the
  ↳ demographic cell.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "ddd_collinearity_diagnostics.R"))

run_hours_ddd_cell_exposure <- function(cleaned_df, exposure_cells,
  cell_cluster_vars = c("GilNK", "TeudaGvocha",
    ↳ "MachozMegurim"),
  controls = DEFAULT_CONTROLS, outcome = "
    ↳ WorkHoursCont") {
  if (!outcome %in% names(cleaned_df)) {
    stop("run_hours_ddd_cell_exposure: outcome column '", outcome, "' is not in
      ↳ cleaned_df.")
  }
  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
  missing_join <- setdiff(exposure_join_vars, names(cleaned_df))
  if (length(missing_join) > 0) {
    stop("run_hours_ddd_cell_exposure: cleaned_df lacks the cell variable(s): ",
      paste(missing_join, collapse = ", "))
  }
  missing_cl <- setdiff(cell_cluster_vars, names(cleaned_df))
  if (length(missing_cl) > 0) {
    stop("run_hours_ddd_cell_exposure: cleaned_df lacks the cluster variable(s): ",
      paste(missing_cl, collapse = ", "))
  }

  n_employed <- sum(cleaned_df$Employed == 1, na.rm = TRUE)
  df_ddd <- cleaned_df %>%
    filter(Employed == 1) %>%
    left_join(exposure_cells, by = exposure_join_vars) %>%
    filter(!is.na(WFH_Exposure))
  n_matched <- nrow(df_ddd)
  message(sprintf(
    "run_hours_ddd_cell_exposure: %d of %d employed rows (%.1f%%) matched a pre-
      ↳ period exposure cell.",
    n_matched, n_employed, 100 * n_matched / n_employed
  ))

  cluster_formula <- as.formula(paste("-", paste(cell_cluster_vars, collapse = "")))
  formula_ddd <- as.formula(paste(
    outcome, "~ Mother * Post * WFH_Exposure +", paste(controls, collapse = " + ")
  ))
  model <- feols(formula_ddd, data = df_ddd, cluster = cluster_formula)
  check_for_dropped_coefficients(model, "run_hours_ddd_cell_exposure()"s triple
    ↳ interaction")

  term <- "Mother:Post:WFH_Exposure"
  if (!term %in% names(coef(model))) {

```

```

    stop("run_hours_ddd_cell_exposure: the triple interaction was not estimated.")
  }

# The regressor's dispersion on the rows the model actually used, for the per-SD
#   ↳ scaling.
used <- df_ddd %>% filter(!is.na(.data[[outcome]]))
reg_sd <- sd(used$WFH_Exposure)
n_clusters <- used %>% distinct(across(all_of(cell_cluster_vars))) %>% nrow()

coefs <- tibble(
  term          = term,
  estimate       = unname(coef(model)[[term]]),
  std_error      = unname(se(model)[[term]]),
  p_value        = unname(pvalue(model)[[term]]),
  n              = as.integer(nobs(model)),
  n_clusters     = n_clusters,
  regressor_sd   = reg_sd,
  estimate_per_sd = unname(coef(model)[[term]] * reg_sd,
  se_per_sd      = unname(se(model)[[term]] * reg_sd
)

table <- etable(model, headers = c(paste0(outcome, " (hours DDD, pre-period cell
  ↳ exposure)")), digits = 4)
print(table)
message(sprintf(
  "run_hours_ddd_cell_exposure: %s = %.4f (SE %.4f) per unit; regressor SD %.4f <->
  ↳ %.4f (SE %.4f) per SD; %d clusters.",
  term, coefs$estimate, coefs$std_error, reg_sd, coefs$estimate_per_sd, coefs$se_
  ↳ per_sd, n_clusters
))

invisible(list(
  table = table,
  model = model,
  coefs = coefs,
  n_employed = n_employed,
  n_matched = n_matched,
  n_clusters = n_clusters,
  regressor_sd = reg_sd,
  cluster_vars = cell_cluster_vars,
  outcome = outcome
))
}

```

robustness/age_balance_robustness.R

```

# age_balance_robustness.R
# Age-balance robustness chain. GilNK (age group) is imbalanced between mothers and
#   ↳ non-mothers
# in the pre-period, and the gap varies by exposure quartile. Two comparison
#   ↳ specifications for
# each DDD: Mother:GilNK added to the formula, and pre-period raking weights that
#   ↳ equalize each
# Mother group's GilNK distribution within exposure quartile. Quartile edges are
#   ↳ computed once on
# the pre-period rows and applied everywhere.
library(tidyverse)
library(fixest)
source(file.path("scripts", "data_processing.R"))
source(file.path("scripts", "occupation_exposure_breaks.R"))
source(file.path("scripts", "assign_wfh_quartile.R"))

compute_pre_period_quartile_breaks <- function(cleaned_df, exposure_cells) {
  # Join keys are derived from exposure_cells' own columns.
  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
  cleaned_df %>%
    left_join(exposure_cells, by = exposure_join_vars) %>%
    compute_occupation_exposure_breaks.caller = "compute_pre_period_quartile_breaks")
}

# 1. Age gap by Mother status, per exposure quartile
diagnose_gilnk_by_quartile <- function(cleaned_df, exposure_cells) {
  gilnk_num <- function(x) as.numeric(as.character(x))
  breaks <- compute_pre_period_quartile_breaks(cleaned_df, exposure_cells)

  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
  pre_df <- cleaned_df %>%

```

```

  filter(ShnatSeker < 2020) %>%
  left_join(exposure_cells, by = exposure_join_vars) %>%
  filter(!is.na(WFH_Exposure)) %>%
  assign_wfh_quartile(breaks)

gap <- pre_df %>%
  group_by(WFH_Exposure_Q) %>%
  group_modify(~ {
    tt <- t.test(gilnk_num(GilNK) ~ Mother, data = .x)
    # Sign flipped to Mother 1 minus 0.
    tibble(
      mean_GilNK_Mother0 = unname(tt$estimate[1]),
      mean_GilNK_Mother1 = unname(tt$estimate[2]),
      gap_Mother1_minus_0 = unname(tt$estimate[2] - tt$estimate[1]),
      t_stat = unname(tt$statistic),
      p_value = tt$p.value,
      n = nrow(.x)
    )
  }) %>%
  ungroup()

message("=== GilNK (age-group) gap by Mother status, per WFH_Exposure quartile (pre
  ↳ -period) ===")
print(gap)

invisible(list(gap_by_quartile = gap, breaks = breaks, pre_df = pre_df))
}

# 2. Employment DDD with Mother:GilNK added
run_ddd_age_interacted <- function(cleaned_df, exposure_cells, controls = DEFAULT_
  ↳ CONTROLS) {
  cell_fe_vars <- c("GilNK", "TeudaGvoha", "MachozMegurim")
  other_controls <- setdiff(controls, cell_fe_vars)
  # Clustered on the cell the regressor varies at.
  cluster_formula <- as.formula(paste("-", paste(cell_fe_vars, collapse = "-"))))

  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
  ddd_df <- cleaned_df %>%
    left_join(exposure_cells, by = exposure_join_vars)

  additive <- feols(
    as.formula(paste("Employed ~ Mother * Post * WFH_Exposure + Mother:GilNK +",
    paste(controls, collapse = " + "))),
    data = ddd_df, cluster = cluster_formula
  )
  # Mother:GilNK is not absorbed by the cell FE.
  fe <- feols(
    as.formula(paste("Employed ~ Mother * Post * WFH_Exposure + Mother:GilNK +",
    paste(other_controls, collapse = " + ")",
    "|", paste(cell_fe_vars, collapse = "-"))),
    data = ddd_df, cluster = cluster_formula
  )

  print(etable(additive, fe,
    headers = c("Age-interacted Spec 1: additive", "Age-interacted Spec 2:
    ↳ cell FE"),
    digits = 4))

  invisible(list(additive = additive, fe = fe))
}

# 3. Raking weights on GilNK within exposure quartile, from the pre-period
build_gilnk_rake_weights <- function(cleaned_df, exposure_cells) {
  breaks <- compute_pre_period_quartile_breaks(cleaned_df, exposure_cells)

  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
  pre_df <- cleaned_df %>%
    filter(ShnatSeker < 2020) %>%
    left_join(exposure_cells, by = exposure_join_vars) %>%
    filter(!is.na(WFH_Exposure)) %>%
    assign_wfh_quartile(breaks)

  # Target: the pooled (both Mother groups) GilNK distribution within each quartile.
  target <- pre_df %>%
    count(WFH_Exposure_Q, GilNK, name = "n_target") %>%
    group_by(WFH_Exposure_Q) %>%
    mutate(p_target = n_target / sum(n_target)) %>%
    ungroup()

  own <- pre_df %>%

```

```

  count(WFH_Exposure_Q, Mother, GilNK, name = "n_own") %>%
  group_by(WFH_Exposure_Q, Mother) %>%
  mutate(p_own = n_own / sum(n_own)) %>%
  ungroup()

weights_tbl <- own %>%
  left_join(target %>% select(WFH_Exposure_Q, GilNK, p_target), by = c("WFH_
  ↳ Exposure_Q", "GilNK")) %>%
  mutate(rake_weight = p_target / p_own)

extreme <- filter(weights_tbl, rake_weight > 10 | rake_weight < 0.1)
if (nrow(extreme) > 0) {
  message(sprintf(
    "build_gilnk_rake_weights: %d of %d (quartile x Mother x GilNK) cells have an
    ↳ extreme raking weight (<0.1 or >10):",
    nrow(extreme), nrow(weights_tbl)
  ))
  print(as.data.frame(select(extreme, WFH_Exposure_Q, Mother, GilNK, n_own, p_own,
    ↳ p_target, rake_weight)))
}

list(weights = weights_tbl %>% select(WFH_Exposure_Q, Mother, GilNK, rake_weight),
  ↳ breaks = breaks)
}

# 4. Hours analogs: occupation-level exposure as regressor, clustered on occupation;
#   ↳ the cell
# quartiles serve only to assign the raking weights.
run_hours_ddd_age_interacted <- function(cleaned_df, exposure_index, controls =
  ↳ DEFAULT_CONTROLS) {
  df_ddd <- cleaned_df %>%
    filter(Employed == 1) %>%
    inner_join(
      exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
      ↳ = wfh_exposure),
      by = "MishlachYad_ISCO_08_2"
    )

  model <- feols(
    as.formula(paste("WorkHoursCont ~ Mother * Post * WFH_Exposure + Mother:GilNK +",
    paste(controls, collapse = " + "))),
    data = df_ddd, cluster = -MishlachYad_ISCO_08_2
  )

  print(etable(model, headers = c("Hours DDD, age-interacted"), digits = 4))

  invisible(list(model = model, n_clusters = n_distinct(df_ddd$MishlachYad_ISCO_08_2)
    ↳ ))
}

run_hours_ddd_reweighted <- function(cleaned_df, exposure_cells, exposure_index,
  controls = DEFAULT_CONTROLS, rake = NULL) {
  if (is.null(rake)) rake <- build_gilnk_rake_weights(cleaned_df, exposure_cells)

  exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))

  # Cell exposure assigns the weight; occupation exposure is the regressor.
  ddd_df <- cleaned_df %>%
    filter(Employed == 1) %>%
    left_join(exposure_cells, by = exposure_join_vars) %>%
    filter(!is.na(WFH_Exposure)) %>%
    assign_wfh_quartile(rake$breaks) %>%
    rename(WFH_Exposure_Cell = WFH_Exposure) %>%
    left_join(rake$weights, by = c("WFH_Exposure_Q", "Mother", "GilNK")) %>%
    inner_join(
      exposure_index %>% select(MishlachYad_ISCO_08_2 = occupation_code, WFH_Exposure
      ↳ = wfh_exposure),
      by = "MishlachYad_ISCO_08_2"
    )

  n_total <- nrow(ddd_df)
  n_missing <- sum(is.na(ddd_df$rake_weight))
  if (n_missing > 0) {
    message(sprintf(
      paste0("run_hours_ddd_reweighted: %d of %d rows (%.1f%%) have no pre-period-
      ↳ derived weight ",
      "for their (quartile, Mother, GilNK) cell -- dropped by feols's listwise
      ↳ deletion."),
      n_missing, n_total, 100 * n_missing / n_total
    ))
  }
}

```

```

}

model <- feols(
  as.formula(paste("WorkHoursCont ~ Mother * Post * WFH_Exposure +", paste(controls
    ↪ , collapse = " + "))),
  data = ddd_df, weights = -rake_weight, cluster = -MishlachYad_ISCO_08_2
)

print(etable(model, headers = c("Hours DDD, reweighted"), digits = 4))

invisible(list(rake = rake, model = model,
  n_clusters = n_distinct(ddd_df$MishlachYad_ISCO_08_2[!is.na(ddd_df$
    ↪ rake_weight)])))
}

run_ddd_reweighted <- function(cleaned_df, exposure_cells, controls = DEFAULT_
  ↪ CONTROLS,
  rake = NULL) {
  if (is.null(rake)) rake <- build_gilnk_rake_weights(cleaned_df, exposure_cells)

  cell_fe_vars <- c("Gilnk", "TeudaGvoha", "MachozMegurim")
  other_controls <- setdiff(controls, cell_fe_vars)

```

```

cluster_formula <- as.formula(paste("-", paste(cell_fe_vars, collapse = "~")))

# Weights are keyed on (quartile, Mother, Gilnk), so pre-period weights apply to
  ↪ every row.
exposure_join_vars <- setdiff(names(exposure_cells), c("WFH_Exposure", "n_cell"))
ddd_df <- cleaned_df %>%
  left_join(exposure_cells, by = exposure_join_vars) %>%
  filter(!is.na(WFH_Exposure)) %>%
  assign_wfh_quartile(rake$breaks) %>%
  left_join(rake$weights, by = c("WFH_Exposure_Q", "Mother", "Gilnk"))

n_total <- nrow(ddd_df)
n_missing <- sum(is.na(ddd_df$rake_weight))
if (n_missing > 0) {
  message(sprintf(
    paste0("run_ddd_reweighted: %d of %d rows (%.1f%%) have no pre-period-derived
    ↪ weight for ",
    "their (quartile, Mother, Gilnk) cell -- dropped by feols's listwise
    ↪ deletion."),
    n_missing, n_total, 100 * n_missing / n_total
  ))
}

```

```

additive <- feols(
  as.formula(paste("Employed ~ Mother * Post * WFH_Exposure +", paste(controls,
    ↪ collapse = " + "))),
  data = ddd_df, weights = -rake_weight, cluster = cluster_formula
)

fe <- feols(
  as.formula(paste("Employed ~ Mother * Post * WFH_Exposure +", paste(other_
    ↪ controls, collapse = " + "),
    "|", paste(cell_fe_vars, collapse = "~"))),
  data = ddd_df, weights = -rake_weight, cluster = cluster_formula
)

print(etable(additive, fe,
  headers = c("Reweighted Spec 1: additive", "Reweighted Spec 2: cell FE
    ↪ "),
  digits = 4))

invisible(list(rake = rake, additive = additive, fe = fe))
}

```